

Instructions for Assemblers

Original Instructions

Company Name: VITTORIO-LUMARE

Product Name: Yeah Hand

Product Type: Robotic Hand

Product in development phase (prototype / quasi-machine) intended only for integrators and beta testers, not for end users.

Manufacturer Address:

Vittorio Lumare

via Boccioni 4

88900 CROTONE (KR)

Italy

Revisions

Version	Comment	Author	Date
1.0	Initial Version	Vittorio Lumare	20/09/2024
1.0 - EN	Translation into English	Fortunato Pantisano	11/11/2025
1.1 - EN	Translation completion (DOI)	Vittorio Lumare	11/11/2025
1.2 - EN	DOI and certificate update	Vittorio Lumare	11/11/2025
1.3 - EN	Added chapter "Residual risks to be communicated to users"	Vittorio Lumare	12/01/2026
1.4 - EN	Updated assembly	Vittorio Lumare	24/06/2026
1.5 - EN	Renamed terms, added terms to the glossary, removed sections related to components no longer part of the product, corrected errors in illustrations	Vittorio Lumare	25/06/2026

1 General warnings and machine compliance

1.1 Document structure

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1.2 Consultation methods

We invite readers to first read the chapter on [safety](#) , which is the most critical and important aspect.

Afterwards, it is possible to skip to the desired chapter, each of which can be consulted on its own, keeping in mind that each chapter refers to the others with some links, so it is possible to consult the relevant section directly.

In the final chapter, there are illustrations with the nomenclature of the parts, which should be consulted during the assembly procedure.

1.3 Legislative references

Directives	Applicable requirements met
Machinery Directive 2006/42/EC	1.1.2, 1.1.3, 1.1.5, 1.3.2, 1.3.3, 1.3.4, 1.3.8.1, 1.5.1, 1.5.2, 1.5.5, 1.5.6, 1.6.1, 1.7.1, 1.7.2, 1.7.4, 1.7.4.1, 1.7.4.2, 4.1.2.3, 4.1.2.4, 4.3.1, Annex VI. It is declared that the relevant technical documentation has been compiled in accordance with Part B of Annex VII of the Machinery Directive. The manufacturer undertakes to transmit, upon request by national authorities, the relevant technical documentation specified in Annex VII, Part B, within the required terms.
2014/30/EU Electromagnetic Compatibility Directive (EMC)	Refer to the EMC directive and the harmonized standards used below.
2011/65/EU Restriction of the use of certain hazardous substances (ROHS)	Refer to the ROHS directive and the harmonized standards used below.

INTERNATIONAL STANDARD	DESCRIPTION
EN ISO 12100:2010	2012 Safety of Machinery — General principles for design — Risk assessment and risk reduction
EN 60204-1	2009 Safety of Machinery – Electrical equipment of machines – Part 1: General requirements
ISO 14539:2000	Manipulating industrial robots — Objects handling with grasping-type grippers — Vocabulary and presentation of characteristics
ISO 14121-2	ISO/TR Safety of Machinery – Risk assessment – Part 2: Practical guidance and examples of methods

1.4 Draft Declaration of Incorporation

Original Declaration of Incorporation

In accordance with Directive 2006/42/EC, Annex II, part B, of the European Parliament and of the Council, relating to machinery,
We, the manufacturer,

VITTORIO-LUMARE
via Boccioni 4, 88900 CROTONE (KR) - Italy

Hereby declare that the following product:

Product Name: VITTORIO-LUMARE Yeah Hand
Description: Electric grasping gripper
Functions: Object manipulation, Pick and Place, gestures.
Models: Yeah Hand L, Yeah Hand R
Types: Robotic grasping gripper, electric and adaptive

Serial number: 2025 0 00000
anno
0:Lato sinistro
1:Lato destro
Sequential numbering, restarting at 0 each year

The partly completed machinery must not be put into service until the final machinery into which it is to be incorporated has been declared in conformity with the provisions of the Machinery Directive. Compliance with all essential requirements of the Machinery Directive depends on the specific robotic application and the overall risk assessment.

Directives	Applicable requirements met
Machinery Directive 2006/42/EC	1.1.2, 1.1.3, 1.1.5, 1.3.2, 1.3.3, 1.3.4, 1.3.8.1, 1.5.1, 1.5.2, 1.5.5, 1.5.6, 1.6.1, 1.7.1, 1.7.2, 1.7.4, 1.7.4.1, 1.7.4.2, 4.1.2.3, 4.1.2.4, 4.3.1, Annex VI. It is declared that the relevant technical documentation has been compiled in accordance with Part B of Annex VII of the Machinery Directive. The manufacturer undertakes to transmit, in response to a reasoned request by the national authorities, relevant information on the partly completed machinery, in accordance with Annex VII, Part B, within the required time limits.
2014/30/EU Electromagnetic Compatibility Directive (EMC)	Refer to the EMC directive and the harmonized standards used below.
2011/65/EU Restriction of the use of certain hazardous substances (ROHS)	Refer to the ROHS directive and the harmonized standards used below.

Person authorized to compile the technical file: Vittorio Lumare. Address: see manufacturer's address.

CROTONE, 07/09/2024 (place and date of issue)



Vittorio Lumare – Technical Director
VITTORIO-LUMARE

The following standards have been applied:

INTERNATIONAL STANDARD	DESCRIPTION
EN ISO 12100:2010	2012 Safety of machinery — General principles for design — Risk assessment and risk reduction
EN 60204-1	2009 Safety of machinery – Electrical equipment of machines – Part 1: General requirements
ISO 14539:2000	Manipulating industrial robots — Object handling with grasping grippers — Vocabulary and presentation of characteristics
ISO 14121-2	ISO/TR Safety of machinery – Risk assessment – Part 2: Practical guidance and examples of methods

1.5 Embedded products

Embedded products

- 5 Servomotors
 - Manufacturer: Shenzhen Waveshare Electronics Co., Ltd
 - Models: ST3215 Servo, ST3215-HS Servo Motor

- Electronic control board
 - Manufacturer: Shenzhen Waveshare Electronics Co., Ltd
 - Model: Servo Driver with ESP32



Figure 1.1: WaveShare Servomotor ST3215-HS



Figure 1.2:
WaveShare ESP32
Servo Driver

1.5.1 Certificates of incorporated products

CERTIFICATE OF CONFORMITY



Certificate No.: CBT240825038EC

Certificate of Conformity

Applicant : **Shenzhen Waveshare Electronics Co., Ltd**
202 2F, World Trade Plaza Funan Community Futian Street, Futian District, Shenzhen

Manufacturer : **Shenzhen Waveshare Electronics Co., Ltd**
202 2F, World Trade Plaza Funan Community Futian Street, Futian District, Shenzhen

Product : **ST3215 Servo**

Model(s) : **ST3215 Servo**
ST3235 Servo, ST3020 Servo, SC09 Servo, WP90 Servo, ST3215-HS Servo Motor, SC15 Servo, ST3025 Servo, RSBL120-24, RSBL45-24, RSBL85-24, RSBL85-12

Trademark : **WAVESHARE**

Test Standard : **EN 55032:2015+A1:2020**
EN 55035:2017+A11:2020
EN IEC 61000-3-2:2019+A1:2021
EN 61000-3-3:2013+A2:2021

The Certificate of Conformity is based on an evaluation of a sample of the above mentioned products. It does not imply an assessment of the whole production. It is possible to use CE marking to demonstrate the compliance with this EMC Directive 2014/30/EU. It is only valid in connection with the test report number: CBT240825038ER.



Shenzhen CABLETEK Compliance Laboratory Limited
East 02, 6th Floor, Building 3, Zhouteng Industrial Zone, Shanglilang Community, Nanwan Street, Longgang District, Shenzhen

Tel: 400 666 1146

E-mail: cbt@cabletek.cn

Web: www.cabletek.cn



Certificate No.: CBT241025045EC

Certificate of Conformity

Applicant : Shenzhen Waveshare Electronics Co., Ltd.
202 2F, World Trade Plaza Funan Community Futian Street, Futian District, Shenzhen

Manufacturer : Shenzhen Waveshare Electronics Co., Ltd.
202 2F, World Trade Plaza Funan Community Futian Street, Futian District, Shenzhen

Product : Servo Driver with ESP32

Model(s) : Servo Driver with ESP32

Trademark : WAVESHARE

Test Standard : EN 55032:2015+A1:2020
EN 55035:2017+A11:2020
EN IEC 61000-3-2:2019+A1:2021
EN 61000-3-3:2013+A2:2021

The Certificate of Conformity is based on an evaluation of a sample of the above mentioned products. It does not imply an assessment of the whole production. It is possible to use CE marking to demonstrate the compliance with this EMC Directive 2014/30/EU. It is only valid in connection with the test report number: CBT241025045ER



Oct. 30, 2024

Shenzhen CABLETEK Compliance Laboratory Limited
East 02, 6th Floor, Building 3, Zhouteng Industrial Zone, Shanglilang Community, Nanwan Street, Longgang District, Shenzhen

Tel: 400 666 1146

E-mail: cbt@cabletek.cn

Web: www.cabletek.cn

Yeah Hand – Adaptive Electric Robotic Hand - Instructions for Assemblers - VITTORIO-LUMARE ©

1.6 Prototype / pre-series label

The final label, including any branding, will be updated in the final version of the product.



2 Safety

2.1 Warnings regarding dangerous situations

2.1.1 Symbol meaning

This paragraph explains the meaning of the symbols used in the rest of the manual.

Please read and understand the meaning of each symbol in this chapter.



READ THE INSTRUCTION MANUAL



OBLIGATION TO USE THE VISE



OBLIGATION TO USE THE MANUAL ASSEMBLY PLIERS



DO NOT SPRAY LIQUIDS ON THE ROBOTIC HAND



DANGER OF FINGER CUTS FROM CABLES



DANGER OF OBJECTS FALLING FROM ABOVE



DANGER OF FLYING OBJECTS



FIRE HAZARD



HOT SURFACE HAZARD



CUTTING HAZARD FOR BODY PARTS



CRUSHING HAZARD FOR BODY PARTS



HAZARD FROM SUBSTANCES HARMFUL TO HEALTH



ELECTRICITY HAZARD

2.1.2 Maintenance operations

!! DO NOT PULL THE TENDONS WITH BARE HANDS !!



MANDATORY

- Use a vise to hold the part being worked on
- Use manual assembly pliers to pull the tendons

POSSIBLE DAMAGE:

- Severe finger cuts
- Bleeding requiring immediate surgical intervention

2.1.3 Commissioning

BEFORE COMMISSIONING THE COMPLETE MACHINE THAT
WILL INTEGRATE THE ROBOTIC HAND, THE FOLLOWING SYMBOLS MUST BE APPLIED ON THE
FINAL PRODUCT, SO THAT THEY ARE ALWAYS VISIBLE TO POSSIBLE PEOPLE
IN THE VICINITY



2.1.4 Typical operation

2.1.4.1 People near the robotic hand



WARNING: the robotic hand could break during use!

POSSIBLE DAMAGE: If the robotic hand breaks, some broken parts could be sharp or pointed and could cut people's skin and cause permanent damage or bleeding and/or require immediate surgery

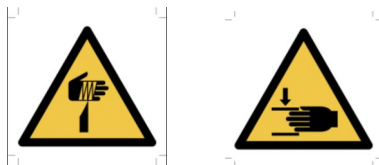
2.1.4.2 Robotic hand holding an object



WARNING: an object could detach from the robotic hand and hit a person

POSSIBLE DAMAGE: A thrown object could cause: broken bones, cuts or impalement burns if it is very hot, contamination if it contains or is composed of harmful substances.

2.1.4.3 Interaction with people



WARNING: human fingers or skin can get stuck in the mechanisms and get injured

POSSIBLE DAMAGE: a finger can be crushed causing severe bruising to the skin or nails, a tendon can cut a human finger causing bleeding and/or require immediate surgery, sharp parts can cut the skin causing loss of blood or bruises.

2.1.5 Risk situations and residual risks

Hazardous Situation	Hazardous Event	Hazard	RESIDUAL RISKS AFTER IMPLEMENTATION OF REDUCTION MEASURES
Handling of tendon wires	Contact of human fingers with the tendon pulled with force	Cutting or abrasion of human fingers by the tendon wires	Skin lesions if the operator does not follow instructions during: • • tendon removal/insertion

2.1.6 Permitted use of the machine and residual risks

Hazardous Situation (ISO 12100:2010, Table B.3)	Hazardous Event (ISO 12100:2010, Table B.4)	Hazard (ISO 12100:2010, Table B.1)	RESIDUAL RISKS AFTER IMPLEMENTATION OF REDUCTION MEASURES
Being near the workspace of the robotic hand (depends on the complete machine integrating the robotic hand)	Breakage of the robotic hand during operation, with exposure of sharp or abrasive parts	Impact, puncture, stabbing, abrasion	On the complete machine, which integrates the robotic hand, produced by others, one of these risks is possible: • A broken part can be sharp and cut human skin until it bleeds and requires immediate surgery.
Being at a sufficient distance to be hit by an object held by the robotic hand in case the object detaches from the robotic hand for any reason	Contact with an object that is ejected/thrown by the robotic hand	Impact, abrasion, cut, stab, puncture, burn, electrocution or contamination by hazardous substances caused by an ejected/ thrown object	On the complete machine, which integrates the robotic hand, produced by others, one of these risks is possible: • A thrown object can cause: bone fractures, cuts or wounds from cutting if sharp, burns if hot, contamination if they contain or are made of harmful substances.
Intensive use of the robotic hand	Flame generated by overheated motors or electronic board	Flame	On the complete machine, which integrates the robotic hand, produced by others, one of the following risks is possible: • The robotic hand can catch fire if used intensively beyond its limits.

2.1.7 Non-permitted use of the machine and residual risks

During the permitted use of the robotic hand, once integrated into the final machine, these hazardous situations can occur:

Hazardous Situation	Hazardous Event	Hazard	RESIDUAL RISKS AFTER IMPLEMENTATION OF REDUCTION MEASURES
Being near the workspace of the robotic hand (depends on the complete machine integrating the robotic hand)	Contact with harmful parts of the gripped object: sharp edges, points, abrasive surfaces	Puncture, stab, abrasion, burn, electrocution	Puncture, stab, abrasion, burn, electrocution
Being near the workspace of the robotic hand (depends on the complete machine integrating the robotic hand)	Contact with a heavy object	Impact	Impact
Being near the workspace of the robotic hand (depends on the complete machine integrating the robotic hand)	Contact with the high-temperature part of the object held in the hand	Burns	Burns
Being near the workspace of the robotic hand (depends on the complete machine integrating the robotic hand)	Contact with the high-voltage part of the object held in the hand	Electrocution, burn	Electrocution, burn
Being near the workspace of the robotic hand (depends on the complete machine integrating the robotic hand)	Contact with a held object contaminated by hazardous substances	Contamination by hazardous substances of part of the held object	Contamination by hazardous substances of part of the held object

It is important that integrators reduce the aforementioned risk situations, as they depend on the final machine and the related risks cannot be reduced by acting solely on the robotic hand.

The following hazardous situations have been analyzed and the related risks have been reduced by the manufacturer, however, they still present residual risks.

Hazardous Situation (ISO 12100:2010, Table B.3)	Hazardous Event (ISO 12100:2010, Table B.4)	Hazard (ISO 12100:2010, Table B.1)	RESIDUAL RISKS AFTER IMPLEMENTATION OF REDUCTION MEASURES
Touching the phalanges of the robotic hand with human fingers	Contact of human fingers with the moving parts of the robotic hand	Crushing of very small human fingers in the space between the robotic finger and the tendon, cutting of human fingers by pulling the tendon with force	On the complete machine, which integrates the robotic hand, produced by others, one of these risks is possible: • The finger can be compressed until it causes severe bruising to the skin or nails • The tendon can cut the human finger until it bleeds and requires surgery
Touching the palm of the robotic hand	Contact of human fingers with the moving parts of the robotic hand	Crushing of human fingers or skin in the space between the base of the thumb and the palm or skeleton	On the complete machine, which integrates the robotic hand, produced by others, one of these risks is possible: • The finger can be compressed until it causes severe bruising to the skin or nails
Being under the robotic hand when holding an object	Falling objects	Impact of a falling object	On the complete machine, which integrates the robotic hand, produced by others, one of these risks is possible: • A falling object can cause: bone breakage, cuts or stabs

Touching an overheated robotic hand	Burn caused by contact with the flame generated by the robotic hand or the robotic hand overheated	Burns	if sharp, burns if hot, contamination if contaminated On the complete machine, which integrates the robotic hand, produced by others, one of these risks is possible: • Skin burns caused by flames
Being at a sufficient distance from a damaged robotic hand to be hit by an object held by the damaged robotic hand in the case where the object detaches from the damaged robotic hand for any reason	Contact with an object ejected/thrown by the damaged robotic hand	Impact, abrasion, cut, stab, puncture, burn, electrocution or contamination by hazardous substances from an object ejected/thrown	On the complete machine, which integrates the robotic hand, produced by others, one of these risks is possible: • A falling object can cause: bone fractures, cuts or lacerations if sharp, burns if hot, contamination if contaminated. • A thrown object can cause: bone fractures, cuts or lacerations if sharp, burns if hot, contamination if it contains or is composed of harmful substances. • A broken part can be sharp and cut human skin causing bleeding and requiring immediate surgery.

2.2 Personal protective equipment

2.2.1 Protective equipment for assembly and maintenance



During the assembly and maintenance phases, it is mandatory to use the following tools:

- Manual assembly pliers to hold and pull the tendons
- Vise to hold the robotic hand steady during the operation

2.2.2 Protective equipment for permitted use

During the permitted use of the robotic hand, no special protective equipment is required.

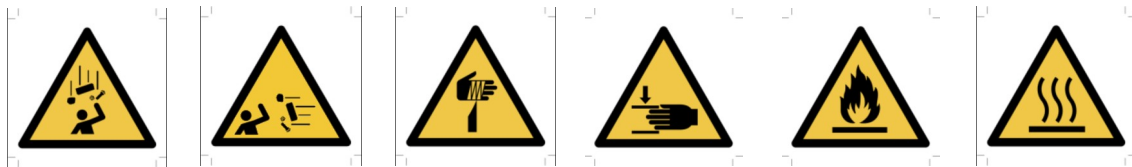
It is the responsibility of the integrators to analyze further risk situations during the use of the complete machine, and to add any appropriate protective equipment.

2.3 Warning plates applied to the machine

These symbols must be applied to the complete machine so that they are visible to the operators of the machine:



These symbols must be applied to the complete machine before commissioning, in a way visible to anyone within the range of the complete machine to anyone, operators or simple passersby or visitors:



These symbols must be applied to the complete machine before commissioning, in a way visible to anyone within the range of the complete machine to anyone, operators or simple passersby or visitors, only in the case that the complete machine integrating the robotic hand is used to manipulate objects containing substances hazardous to health



2.4 Emergency situations

In case of fire, disconnect the power supply.

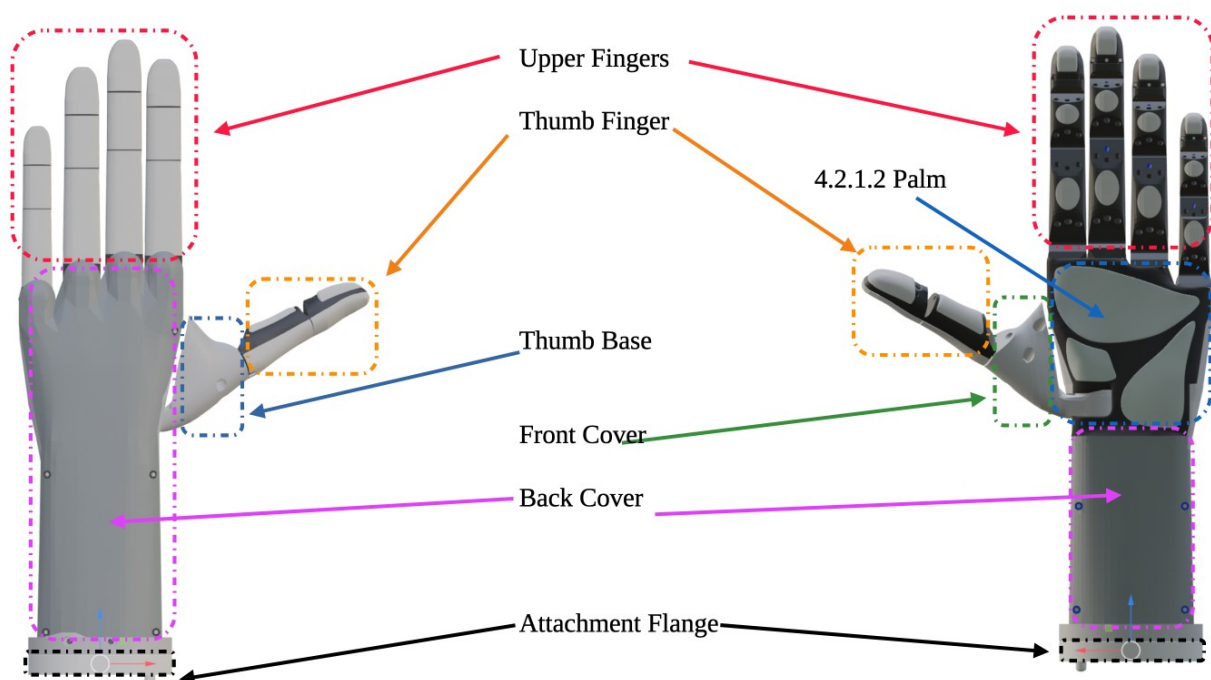
In case of contamination by hazardous substances, the integrators who manufactured the complete machine that manipulates such substances must perform a risk analysis and evaluate the emergency situations and write them in the technical file and in the instructions of the complete machine.

3 Machine overview

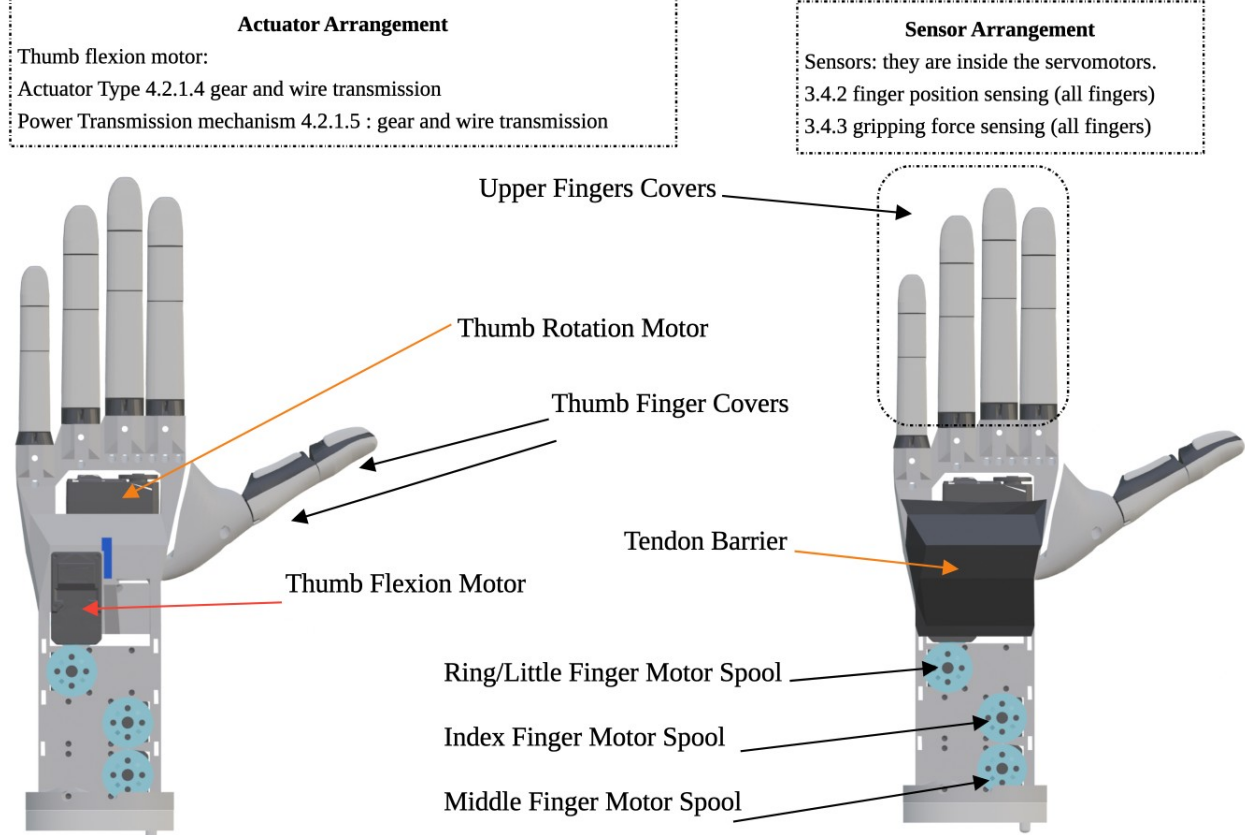
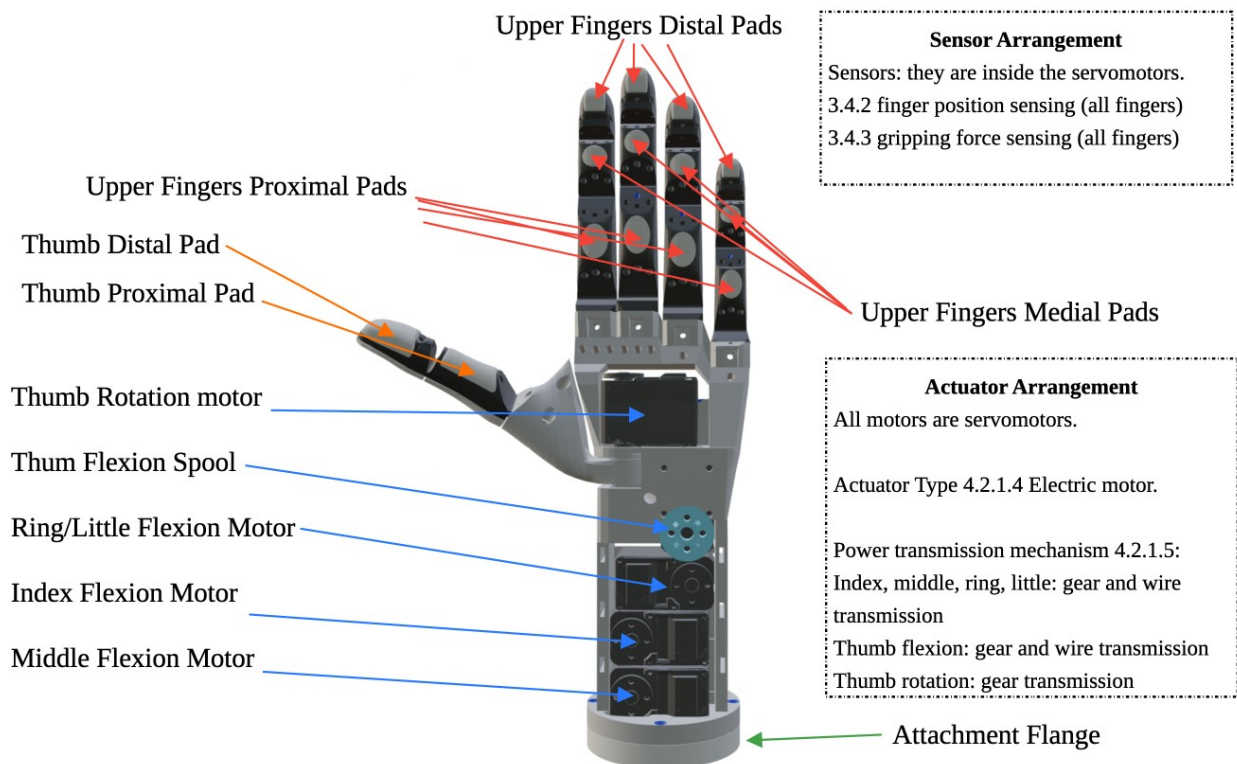
3.1 Machine description

Manufacturer	VITTORIO-LUMARE
Type of Machinery	Robotic Hand (QUASI MACHINE)
Hand Model	Yeah Hand L (Left Hand), Yeah Hand R (Hand Right) The left-side model resembles the "left" human hand. The right-side model resembles the "right" human hand.
Type of robotic hand	Adaptive Electric Robotic Hand
Main applications of the robotic hand	Service robotics: • Grasp/release objects • - Pick and place • Use of panel commands • Use of mechanical buttons • Hand gestures for human-robot interactions

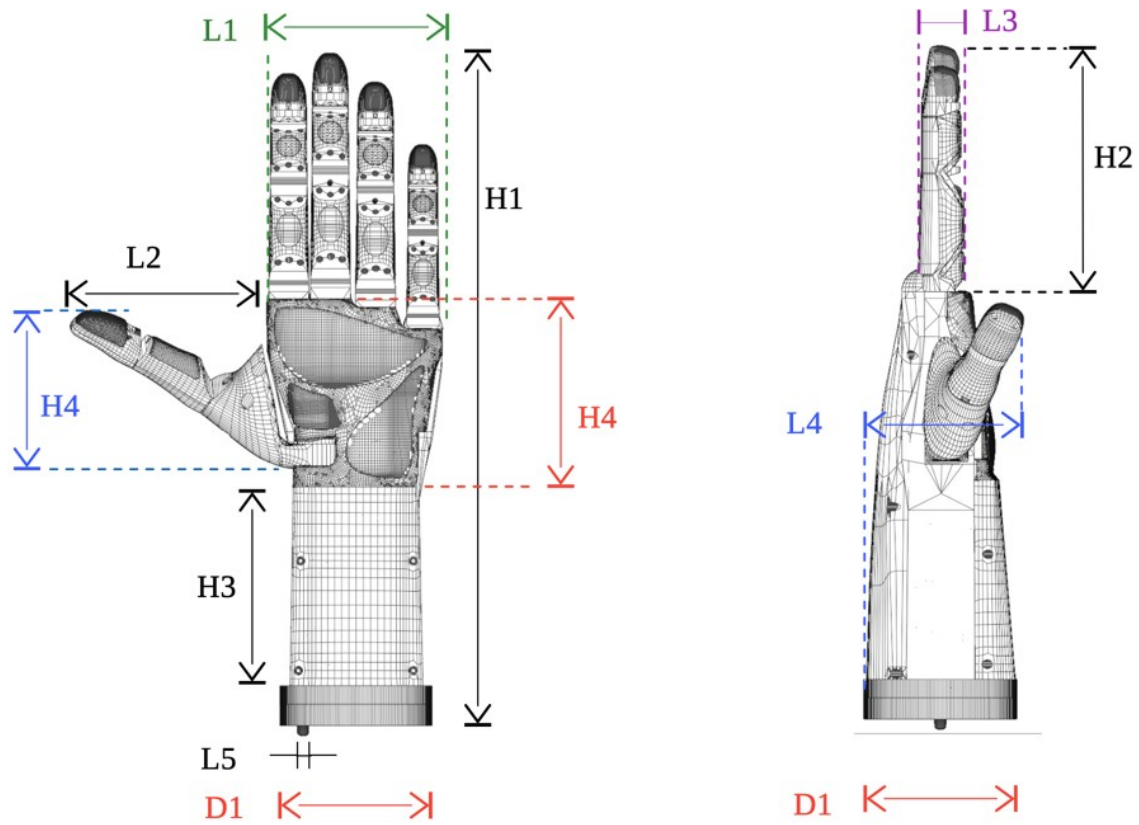
Overall Technical Drawing - External View



Detailed Technical Drawing - Internal View



Detailed Technical Drawing - Measures



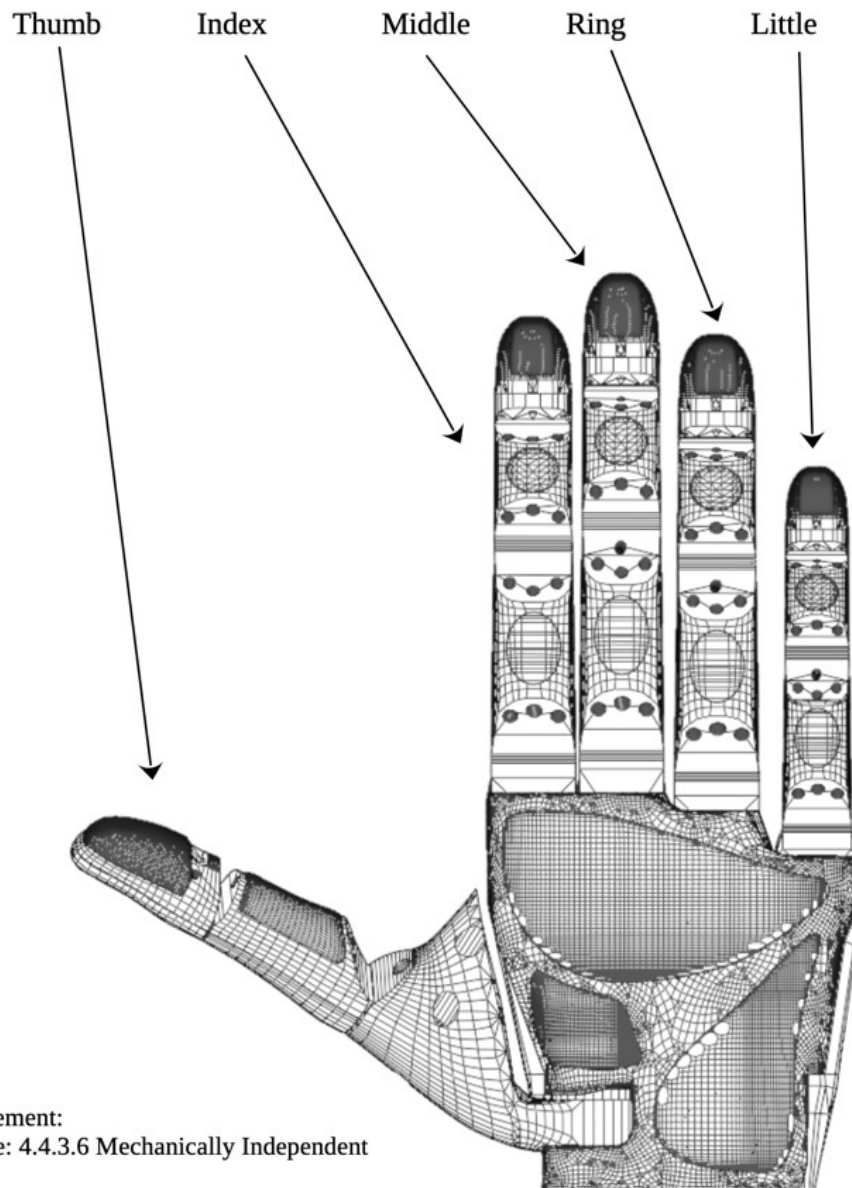
Measures

L1 : 83 mm
 L2 : 99 mm
 L3 : 21 mm
 L4 : 74 mm
 L5 : 5.5 mm
 D1 : 71 mm
 H1 : 318 mm
 H2 : 117 mm

Tolerance : ± 0.4 mm

Detailed Technical Drawing

Finger Arrangement



Type of Movement:

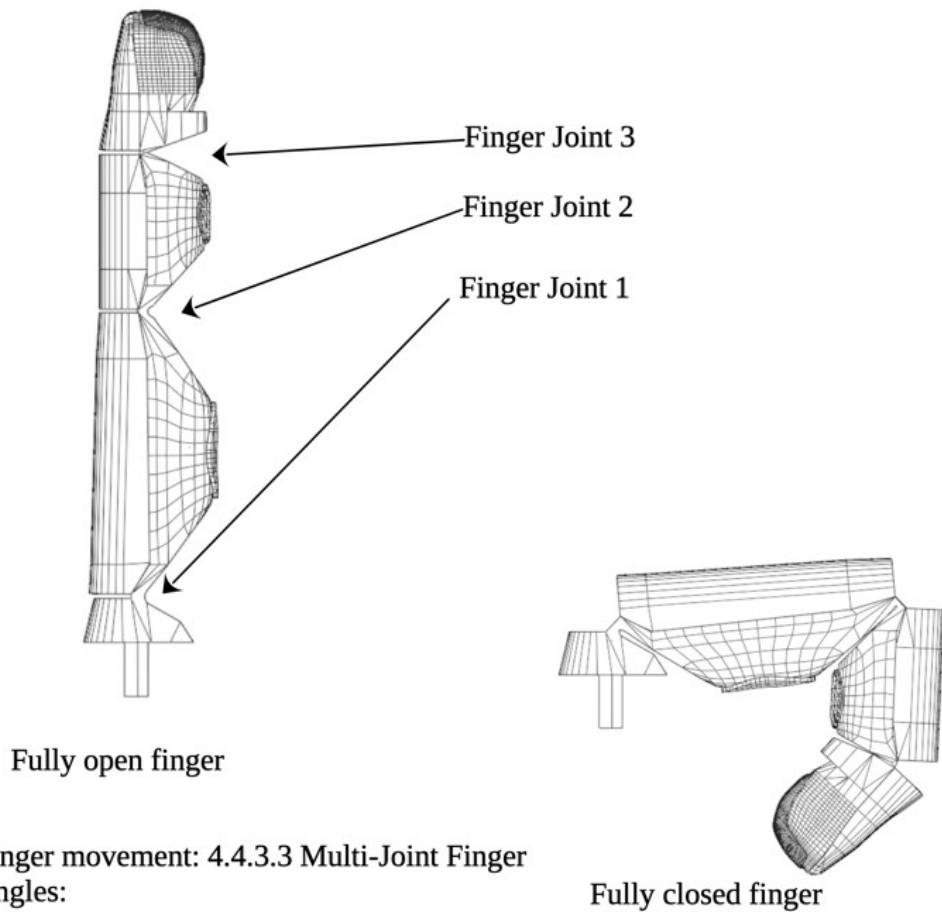
Index, Middle: 4.4.3.6 Mechanically Independent
Fingers

Ring, Little: 4.4.3.5 Mechanically Interrelated
Fingers (Ring and Little fingers move together)

Detailed Technical Drawing

Upper Fingers Movement
It applies to Index, Middle, Ring and Little fingers

Flexion Angle: 70 degrees



Type of finger movement: 4.4.3.3 Multi-Joint Finger

Flexion angles:

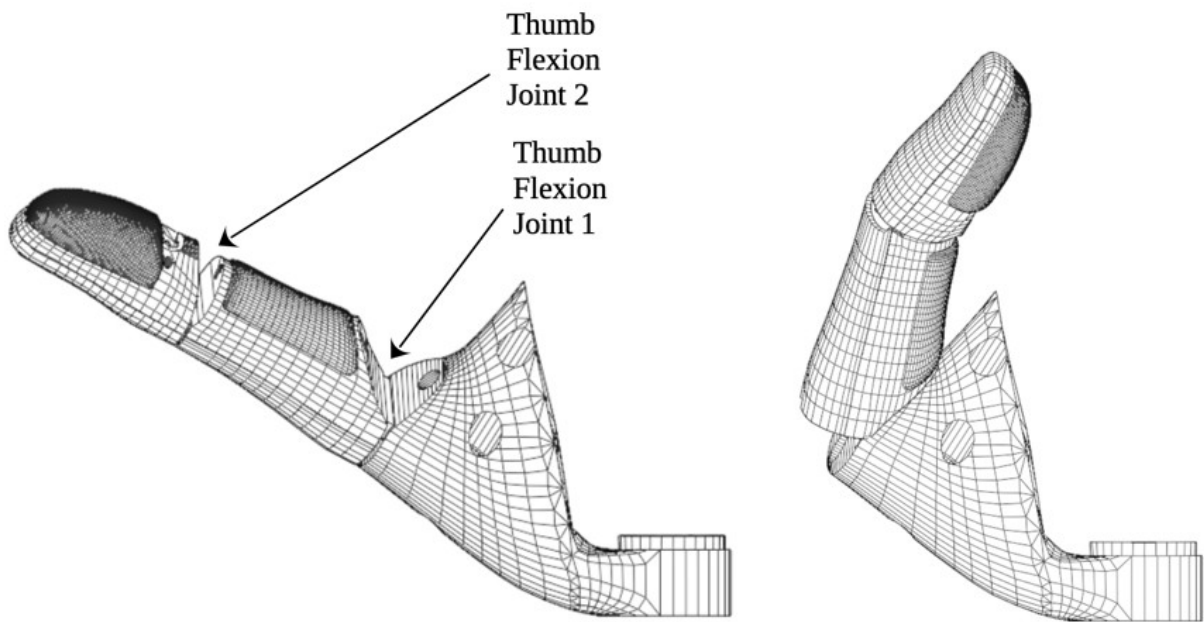
Finger Joint 1 : 87 degrees

Finger Joint 2 : 100 degrees

Finger Joint 3 : 43 degrees

Detailed Technical Drawing

Thumb Finger Movement : Flexion



Type Of Finger Movement:

4.4.3.3 Multi-Joint Finger

4.4.3.6 Mechanically Independent finger

Flexion Angles

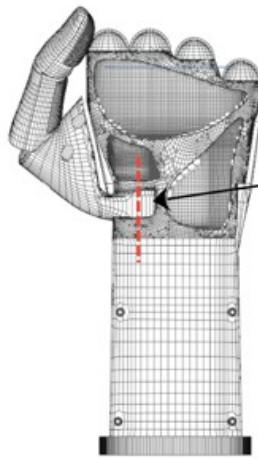
Thumb Flexion Joint 1 : 70 degrees

Thumb Flexion Joint 2 : 25 degrees

Thumb Finger Movement : Rotation
Full Hand View

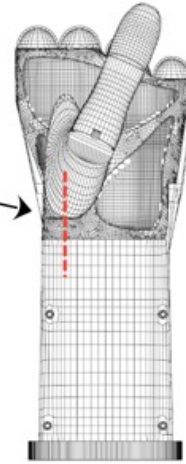
Fully Open

Fully Closed

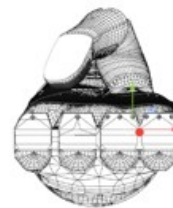
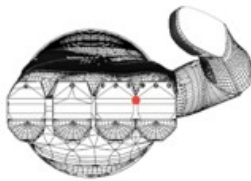
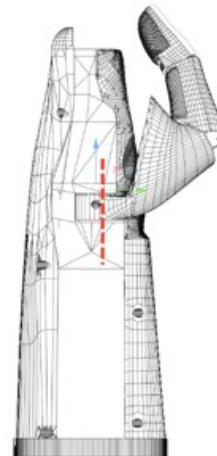
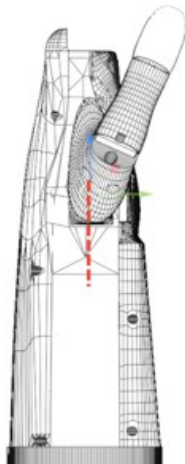


Thumb Rotation Joint

Rotation axis in red color



Note: upper fingers are hidden in this view



Type Of Finger Movement: 4.4.3.3 Multi-Joint Finger
Rotation angle:
Thumb Rotation Joint : 79 degrees

3.2 Intended use

This electric robotic hand is adaptive and is intended for robotic applications.

The robotic hand must be installed on a handling device, such as:

- a mobile platform
- a robotic manipulator
- a humanoid service robot
- a non-humanoid service robot

The robotic hand is suitable for applications:

- educational:
 - schools
 - universities
- research:
 - laboratories
 - research centers
 - universities
- prototyping:
 - development of service robots (humanoid or otherwise) that integrate the robotic hand
- commercial
 - integrated into service robots:
 - in public areas
 - for domestic use
- hobbyist

The robotic hand is not suitable for industrial applications, as it does not possess sufficient:

- robustness
- repeatability
- high immunity to disturbances typical of the industrial environment

3.3 Reasonably foreseeable misuse

Situations of misuse:

- coming into physical contact with living beings
- manipulation of skin or body parts
- shaking hands with people or animals
- highly humid environment or with the presence of water droplets or high-density vapors
- temperatures above 50 °C
- lifting objects heavier than 2kg
- handling substances hazardous to health, radioactive, highly hot (over 50 degrees)
- handling liquids or powdery substances or with small particles (smaller than 12 mm)

3.4 Technical Characteristics of the machine

Robotic hand performance	
Robotic hand mass (kg)	650g without covers / 750g with covers
Maximum distance between fingers (mm)	160 mm
Minimum distance between fingers (mm)	0
Maximum gripping force (N)	90N (TBD : calculated)
Maximum manipulation force (N)	35N (TBD : calculated)
Maximum allowable finger moment (N*m)	0.1
Gripping surface (mm ²)	1260
Operating time: Grip (s)	2
Operating time: Release (s)	2
Type of robotic hand	
Number of Fingers	5
Finger arrangement	Humanoid
Robotic hand mobility degrees (4.3.1)	Spatial grip: 6
Robotic hand degrees of freedom (4.3.2)	5
Grip type (4.3.6)	4.3.6.2 non-centric grip 4.3.6.3 adaptive grip 4.3.6.5 asymmetric grip 4.3.6.6 grip force
Fingers	
Finger mobility degrees (4.4.1)	Index: 3 Middle: 3 Ring: 3 Little: 3 Thumb: 5
Finger degrees of freedom (4.4.2)	Index: 1 Middle: 1 Ring + Little: 1 Thumb: 2
Finger movement (4.4.3)	- Index: - multi-joint fingers (4.4.3.3) - mechanically independent fingers (4.4.3.6) - Middle: - multi-joint fingers (4.4.3.3) - mechanically independent fingers (4.4.3.6) - Ring: - multi-joint finger (4.4.3.3) - mechanically interconnected finger (4.4.3.6) - Little: - multi-joint finger (4.4.3.3) - mechanically interconnected finger (4.4.3.6) - Thumb: - multi-joint fingers (4.4.3.3) - mechanically independent fingers (4.4.3.6)

Actuators (4.2.1.4)	
Type of actuators	Electric motors
Number of actuators	5
Nominal actuator power (W)	60W
Power transmission mechanisms (4.2.1.5)	
Finger Mechanism (index, middle, ring, little finger)	Gear + cable
Thumb flexion mechanism	Gear + cable
Thumb rotation mechanism	Gear
Note: "cable" is the tendon, it is a braided polyethylene fishing line	
Finger movement details (referring to the mechanical interface coordinate system as defined in ISO 9787 standard):	
TCP definition (3.3.8): Offset from plane XmYm	Zoffset = 146.1 mm
TCP definition (3.3.8): Offset from axis Zm	Xoffset = -0.1 mm Yoffset = 18.9 mm
Finger Control	
Finger control scheme (4.5)	4.5.6 Hybrid Control
Control signals	Wireless serial character commands (BT, WiFi) - Signal 1 : Select Mode - Tendon Installation - Tendon Calibration - Relaxation (no movement) - Grip control (POWER, POWERSMALL, MONKEY, PINCH) - RAW control (full control of each finger joint) - Signal 2: - Finger Closed/Open: 0-100 %
Feedback signals	Wireless serial characters (BT, WiFi) - Position: integer value - Speed: integer value - Load: integer value - Temperature: integer value - Current: integer value - Movement: integer value (0: false, 1: true) - Status: integer value
Sensing functions (3.4)	
Position sensing	Yes: Resolution: 4096, Range: [0, 360] degrees
Torque sensing	Yes: Resolution: 2000, Range: [-25, 25] Kg*cm
Temperature sensing	Yes: Precision: 1°C, Range: [0, 70]°C
Voltage sensing	Yes: Precision: 0.1V, Range: [0, 12]V
Current sensing	Yes (TBD)
Movement sensing	Value: 1: Movement, 0: Constant
Clamping elements (4.6)	
Material	- Rigid polymer (e.g. ABS / PLA / PETG / nylon) - Flexible polymer (e.g. TPU / TPE) - Skin-safe silicone (e.g. Smooth-On Dragon Skin NV / Reschimica R PRO GLASS)

Replaceable	Yes	
Interchangeable	Yes	
Operating environment		
Temperature range	10 – 40 °C	
Relative humidity range	0 - 80	
IP protection	IP20 (INSUFFICIENT, MUST BE >IP4X)	
Object properties		
Geometry (shape, dimensions) (mm)	Shape: Any shape Dimensions: maximum diameter 10 cm	
Object mass (kg)	0 - 2 kg TBD	
Maximum/minimum object temperature (°C)	10°C - 40°C	
Surface characteristics	Anything that does not present: sharp edges/points, abrasive surfaces, harmful substances, liquids, radioactive materials	
Magnetic properties	< 1T	
Robotic interfaces (4.7) and wrist mechanisms:		
Mechanical interface	Compliant with standard	UNI EN ISO 9409-1:2005
	Designation code	ISO 9409-1-50-4-M6
Functionality (power and signal transmission)	Power transmission: direct current 12 V, max 14 A No signal transmission via the robot interface (transmission is wireless)	
Warning		
Risk (in the final machine) of: crushing, puncture, cutting, stabbing, fire, burn		

3.5 Commands and signals

The robotic hand does not include physical controls, it is equipped with a wireless serial Bluetooth character interface.

The implementation of any commands is the responsibility of the complete machine integrators.

3.6 Environmental limits of the machine

Operating environment	
Temperature range	10 - 40 °C
Relative humidity range	0 - 80
IP protection	TBD: IP40

4 Transport, handling, storage

4.1 Transport

The robotic hand can be transported by hand.

It is recommended to transport the robotic hand in packaging that prevents:

- impact with surfaces that could compromise the structural integrity or the finish of the external surfaces.
- the infiltration of liquids or dust or materials with dimensions smaller than 12 mm TBD

It is recommended to disassemble the robotic hand from the machine on which it is integrated if it is being transported, as it could be damaged during transport.

During transport, the temperature must not exceed 70 °C, otherwise geometric alterations of the materials it is composed of could occur.

IMPORTANT: Before transporting the robotic hand, disconnect the power supply!

4.2 Handling

It is recommended to use a robotic manipulator to handle the robotic hand.

4.3 Storage

The robotic hand can be stored under these conditions:

- temperature
 - 5°C - 40°C
- humidity:
 - 0 - 70% RH
- light:
 - store in the dark in a closed box
 - do not expose to UV rays

5 Assembly

These instructions explain how to assemble the robotic hand from its pre-manufactured components.

5.1 Servo motor installation and registration

This section describes how to install the servo motors into the hand skeleton and how to register them on the logic board to ensure proper operation.

The procedures are intended for adult users with basic skills in electronics and mechanical assembly.

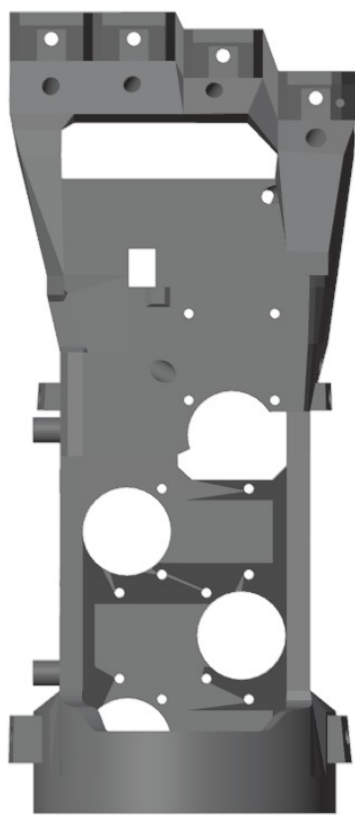


Figure 5.2: Left Hand - Skeleton - Front View – no components mounted

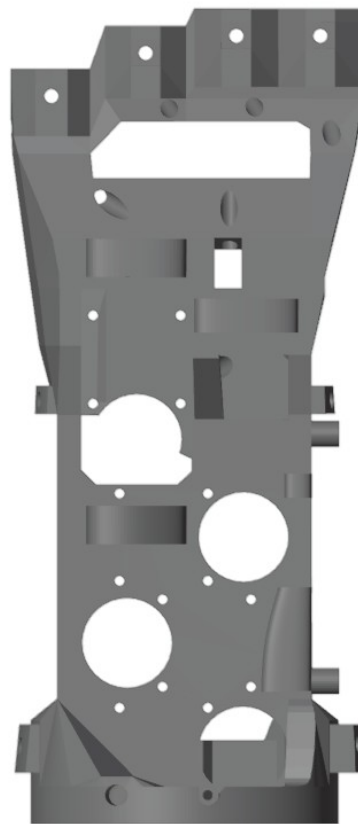


Figure 5.1: Left Hand - Skeleton - Rear View – no components mounted

5.1.1 Basic Thumb and Thumb Rotation servomotor installation (20 min)

Note: TRS is an acronym for Thumb Rotation

Servo Installation:

1. Install Thumb Base on the TRS flange (4 M3 bolts, 16 mm long) as shown in figure 5.3.
2. Install TRS on the TRS Platform as shown in figure 5.4 and screw with 4 M3 self-tapping screws, max 6 mm long (DO NOT use longer screws otherwise the servomotor will break)
3. Insert the assembled TRS Platform+Thumb Base onto the skeleton as shown in figure 5.5 and screw with 4 M2 bolts, 5 mm long.



Figure 5.3: Thumb Base on the Thumb Rotation Servomotor

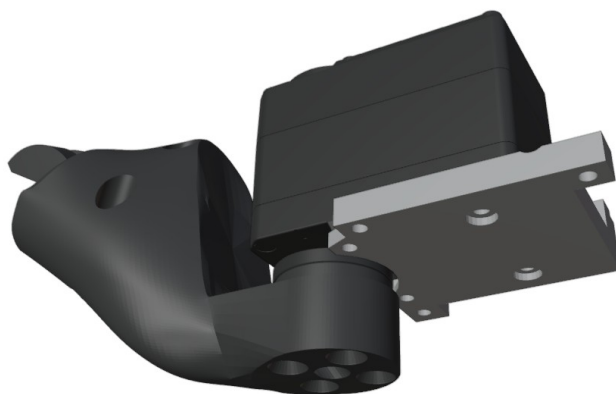


Figure 5.4: Thumb Base and TRS on TRS Platform

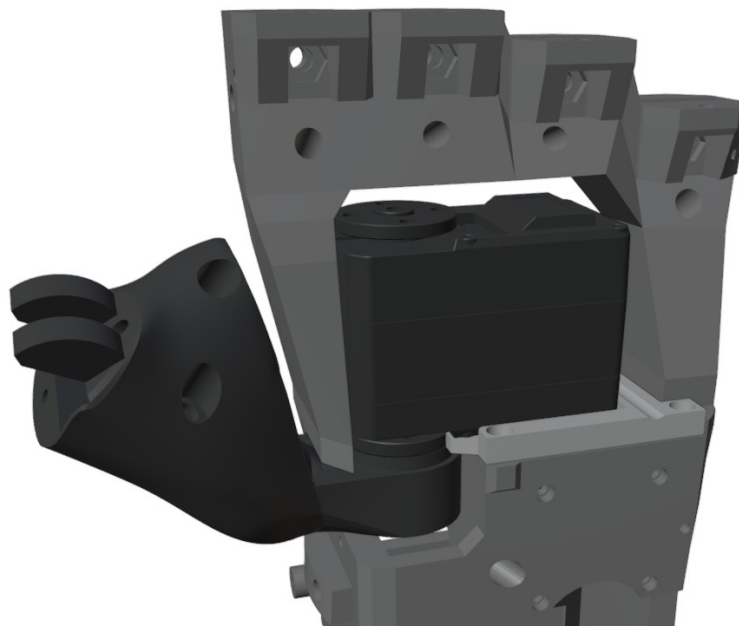


Figure 5.5: TRS and TRS Platform installed on the skeleton

5.1.2 Installation of 4 servomotors for finger flexion (15 min)

Note: each flexion servomotor (there are 4) must be fixed in its slot using 4 self-tapping M2 screws 6-8 mm long.

1. Install each servomotor in its relative slot as shown in figures 5.5 and 5.6
2. Screw the servomotor to the skeleton holes using the four screws provided.
3. ATTENTION: Use the provided screwdriver and tighten the screws slightly to avoid damaging the threaded seat of the servomotor, which is very delicate.
 1. Note: before installing the last servo at the bottom (the one at the base of the skeleton), connect 2 control cables to it (and only to it).
 2. If you have difficulty inserting the last motor at the bottom with the cables, it is recommended to remove the circular aluminum flange, insert the wired motor into the skeleton, screw the servomotor to the skeleton and only finally mount the flange on the servo.
4. Ensure that all servomotor control cables are disconnected except for those of the lowest servo.

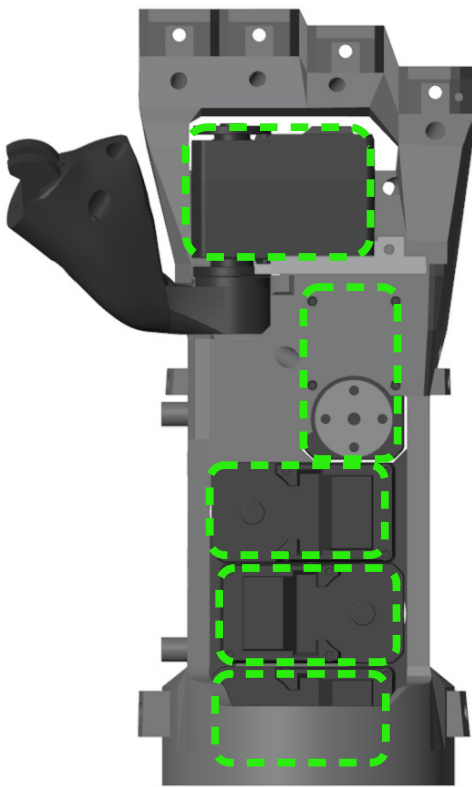


Figure 5.6: Mounted flexion servomotors - front view

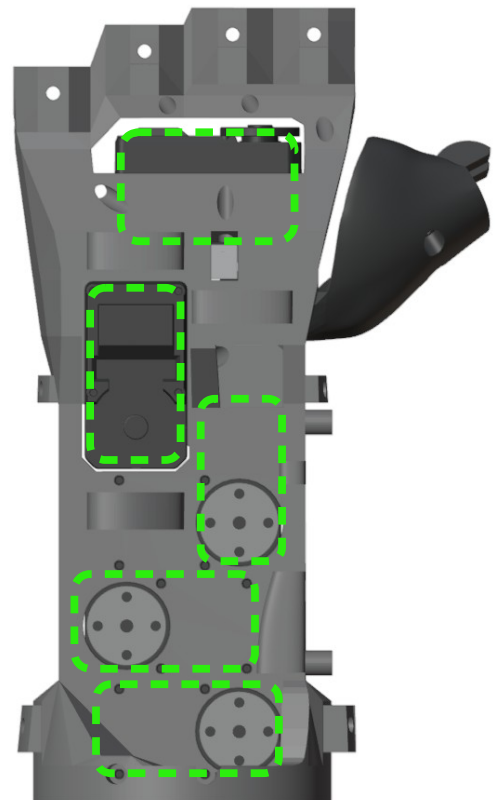


Figure 5.7: Mounted flexion servomotors - rear view

5.1.3 Servomotor registration and calibration

Before starting:

- Ensure the logic board is
 - removed from the hand
 - turned off and disconnected from both the power supply and the USB cable.
- Disconnect all servomotors from the logic board.

Procedure (for each flexion servomotor):

1. Connect the logic board to the PC via USB cable.
2. Upload the program corresponding to the servomotor (table 5.1).
3. Once the upload is complete, disconnect the USB cable .
4. Important : never power the board with USB connected and external power supply simultaneously.
5. If the servomotor is for flexion (finger bending via tendons):
 1. Screw a bolt into one of the holes of the circular plate.
 2. Rotate the plate until the hole with the bolt is at the bottom right (figure 5.9).
6. If the servomotor is for thumb rotation:
 1. Position the thumb at approximately the middle of the range (figure 5.8).
7. Connect the servomotor to the logic board (leave all others disconnected).
8. Power the logic board with the power connector (not with the USB cable!)
9. The program automatically executes:
 1. registration of the servomotor ID
 2. registration of the center position of the range
 3. 4 oscillations of the plate at the end of the procedure (to confirm successful outcome)
10. Important: Turn off the logic board by disconnecting the power supply.
11. Important: Disconnect the servomotor from the logic board.

TABLE 5.1: Servomotor / Registration program (for the logic board)

Servomotor	Registration program
Thumb Rotation	Thumb Rotation Registration Program
Thumb Flexion	Thumb Flexion Registration Program
Index Flexion	Index Flexion Registration Program
Middle Flexion	Middle Flexion Registration Program
Ring Flexion	Ring Flexion Registration Program

Note: The little finger servomotor is not present because it is managed by the ring finger motor.

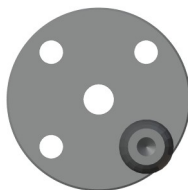


Figure 5.9: Flexion servomotor registration

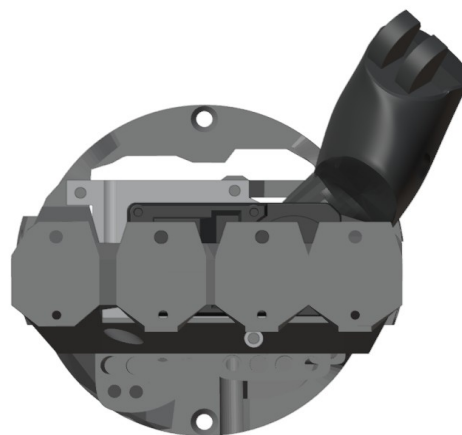


Figure 5.8: Thumb rotation servomotor registration

5.2 Logic board programming with the operating program

Once the servomotors are registered, it is necessary to program the board with the operating program of the robotic hand. The program is 'RebeliaRobot'.

1. DISCONNECT THE SERVOMOTORS FROM THE BOARD
2. Open Arduino IDE
3. Select the board: ESP32-WROOM-DA Module
4. Open the RebeliaRobot.ino program
5. Disconnect the power connector from the logic board
6. Connect the logic board to the PC with the USB cable
7. Upload the program from Arduino IDE
8. Wait for the operation to complete
9. Disconnect the USB cable from the logic board
10. Connect the power connector to the logic board

5.2.1 Servomotor connection

The servomotors are connected to the logic board in a daisy chain, see figures 5.10 and 5.11.

1. Connect the logic board to servomotor 2
2. Connect servomotor 2 to servomotor 3
3. Connect servomotor 3 to servomotor 1
4. Connect servomotor 1 to servomotor 4, first inserting the cable into cavity S1 and letting it exit from cavity S2
5. Connect servomotor 4 to servomotor 5, inserting the cable into cavity S3

At this point, all servomotors are connected to the logic board in a daisy chain.

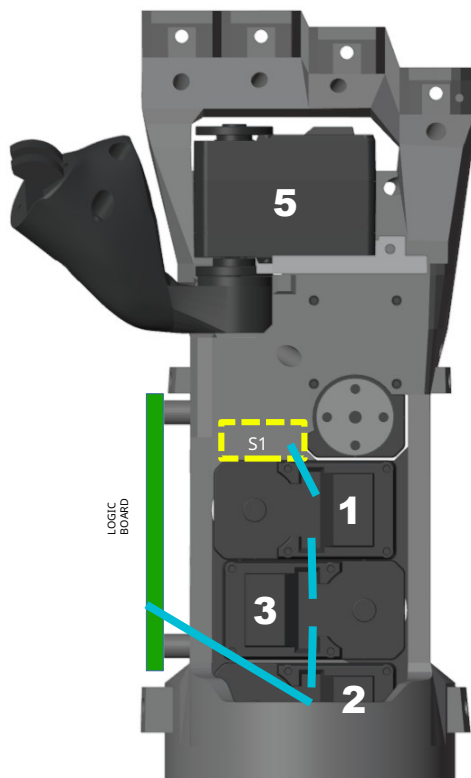


Figure 5.10: Front view. Servomotor IDs, logic board and cavity S1

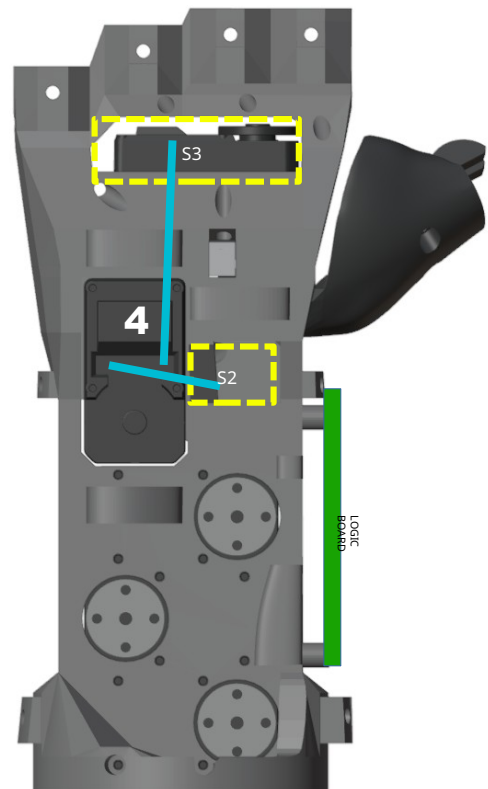


Figure 5.11: Rear view. Servomotor IDs, logic board and cavity S2

5.3 Logic board installation

1. Install the logic board onto the skeleton as shown in figure 5.12, using 4 M2 6 mm long self-tapping screws.

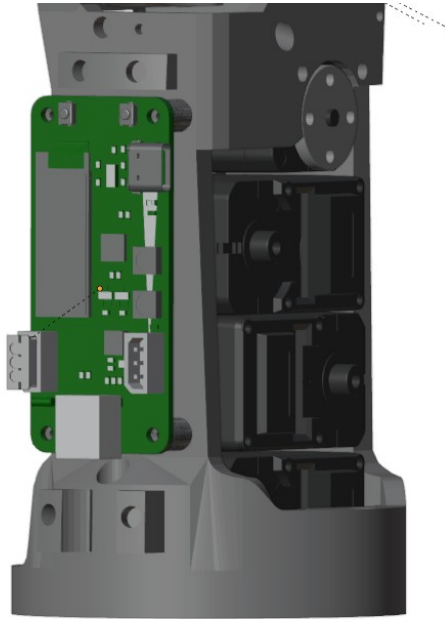


Figure 5.12: Logic board installed

5.4 Finger tendon installation

5.4.1 Upper finger tendon installation



WARNING: Tendons can cut fingers!

MANDATORY: Use the vise to secure the skeleton and the manual assembly pliers to pull the tendon.

Index finger tendon installation (the same procedure also applies to the middle, ring, and little fingers):

Note: At the moment, the assembly KIT is supplied with the tendons already installed on the fingers.

5.4.2 Thumb tendon installation



WARNING: Tendons can cut fingers!

MANDATORY: Use the vise to secure the skeleton and the manual assembly pliers to pull the tendon.

Note: At the moment, the assembly KIT is supplied with the tendons already installed on the fingers.

5.5 Index, Middle, Ring, and Little finger installation

Note: each finger must be fixed, in its slot on the skeleton, with an M3 10mm bolt.

1. Insert the finger into its slot on the skeleton (figure 5.13)
 1. leave the tendons out, they will be inserted into the skeleton holes later in another procedure
2. Place the nut in its seat on the front of the skeleton (figure 5.14)
3. Secure the nut with the provided insertion tool (figure 5.16)
 1. Note: the tool also serves to keep the nut steady in its hexagonal seat while screwing the bolt
4. Insert the bolt into the hole on the back of the skeleton and screw it in (figure 5.15)

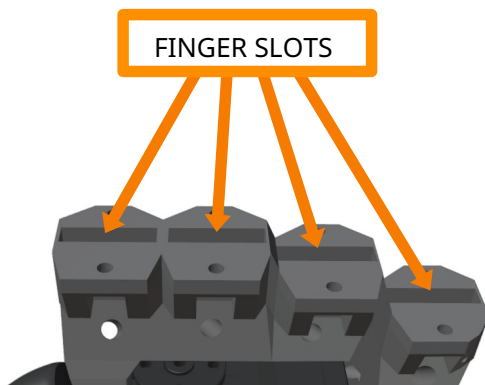


Figure 5.13: Finger slots

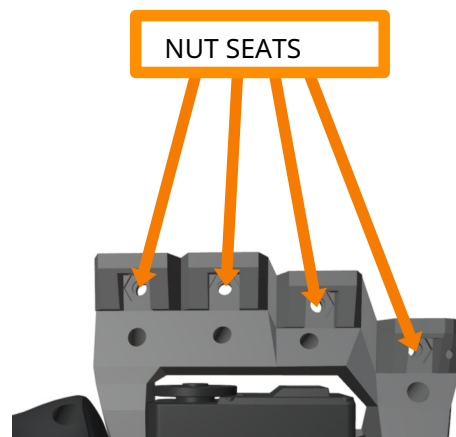


Figure 5.14: Nut seats

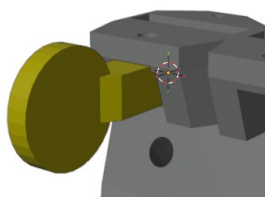


Figure 5.16: Nut insertion and holding tool

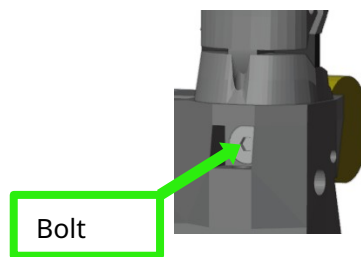


Figure 5.15: Finger bolt insertion

5.6 Thumb finger installation

Note: the thumb must be fixed, in its slot on the thumb base, with two M3 countersunk head bolts, one 16mm long and one 20mm long.

1. Insert the finger into the slot (figure 5.17)
2. Insert two bolts from behind (figure 5.17)
 1. Bolt 1: M3 countersunk head 16 mm long
 2. Bolt 2: M3 countersunk head 20 mm long
3. Secure with two M3 nuts placed on the opposite side (figure 5.18)

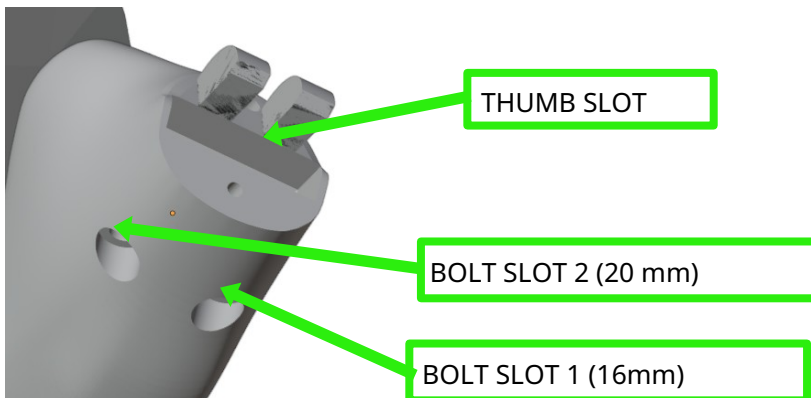


Figure 5.17: Thumb slot and bolt slots

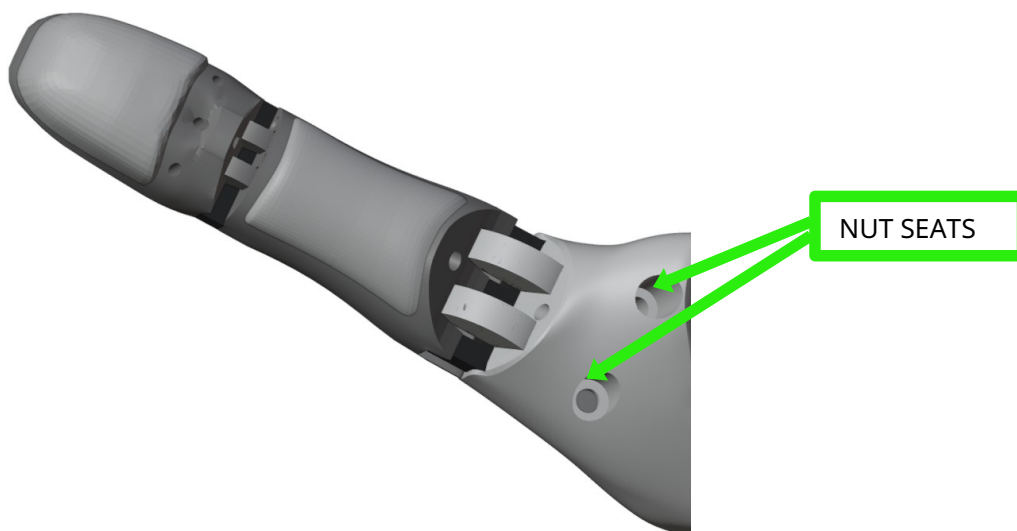


Figure 5.18: Thumb fixing with nuts

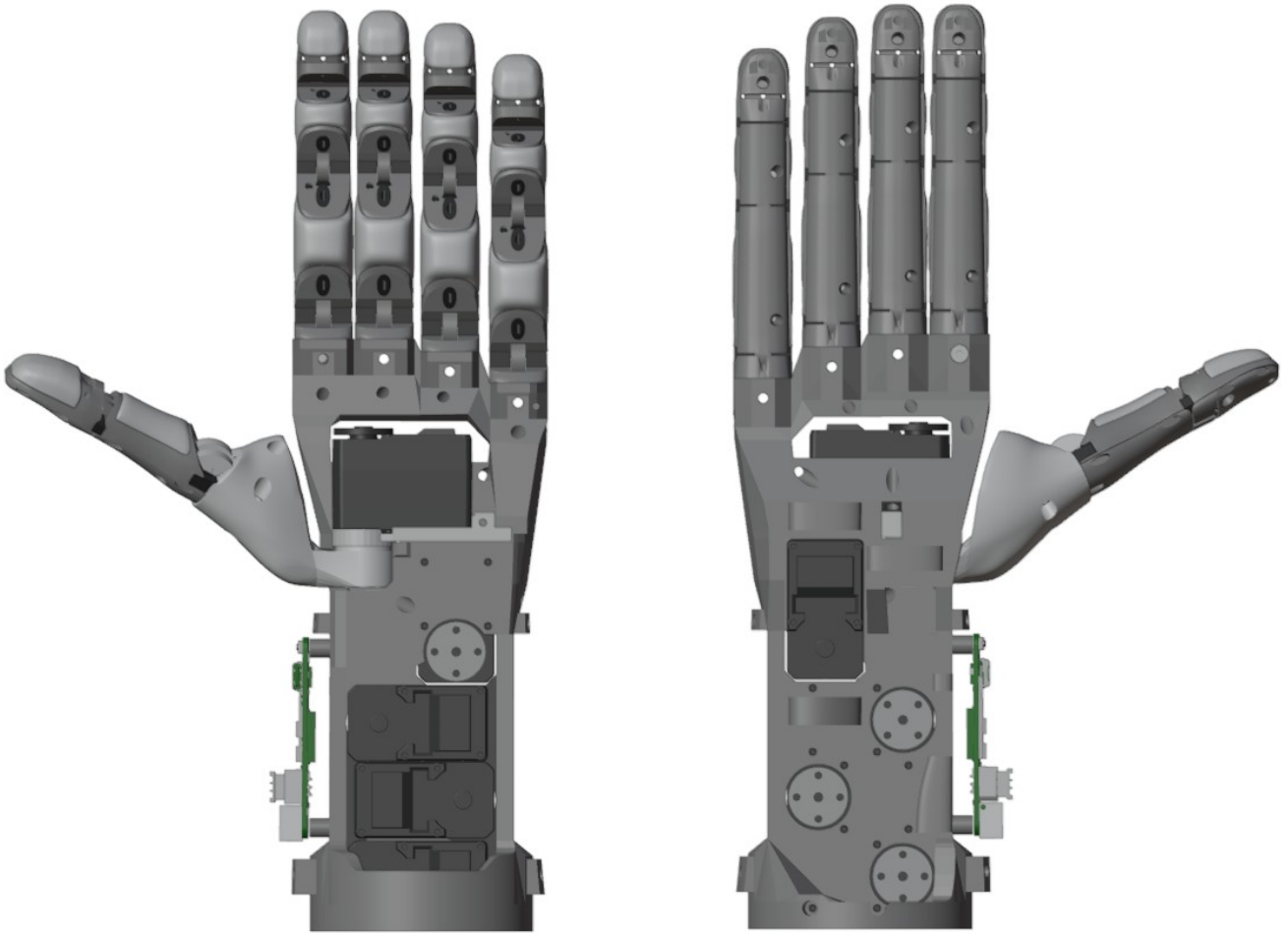


Figure 5.19: Fingers installed (front/back view)

5.7 Index finger PTFE tube installation

For the INDEX finger only:

1. Flexion tendon:
 1. Insert the PTFE tube into the exit hole (on the back of the skeleton) of the flexion tendon (figure 5.20), pull it out from the entry hole on the front of the skeleton and insert it into the initial slot figure 5.22)
 2. Take the remaining part (on the back of the skeleton) and insert it into the final slot shown in figure 5.21 .
 3. Note: if the tube is too long, shorten it by cutting it with a utility knife.
2. Extension tendon:
 1. Insert one end of the PTFE tube into the initial slot (on the back of the skeleton) of the extension tendon (figure 5.24)
 2. Insert the other end into the final slot shown in figure 5.23 : if the tube is too long, shorten it by cutting it with a utility knife.

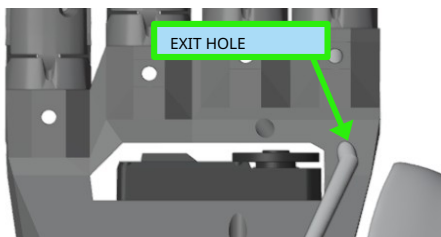


Figure 5.20: Inserting tube into the hole of tendon exit (rear view)

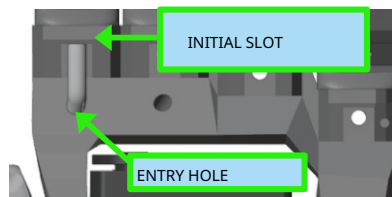


Figure 5.22: Inserting tube into the initial slot (front view)

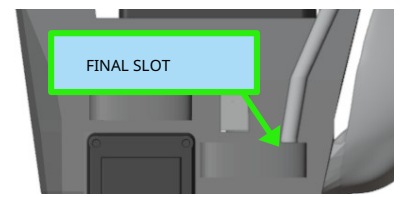


Figure 5.21: Inserting tube into the final slot (rear view)

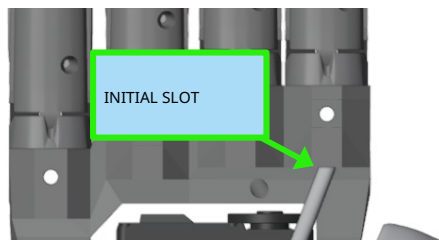


Figure 5.24: Inserting tube into the initial slot (rear view)

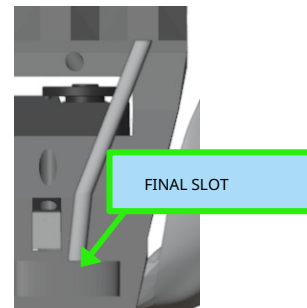


Figure 5.23: Inserting tube into the final slot (rear view)

5.8 Middle finger PTFE tube installation

For the MIDDLE finger only:

1. Flexion tendon:
 1. Insert the PTFE tube into the exit hole (on the back of the skeleton) of the flexion tendon (figure 5.26), let it come out from the entry hole on the front of the skeleton and insert it into the initial slot figure 5.27)
 2. Take the remaining part (on the back of the skeleton) and thread it first through the 2 passage tunnels and then into the final slot figure 5.25).
 3. Note: if the tube is too long, shorten it by cutting it with a utility knife.
2. Extension tendon:
 1. Insert one end of the PTFE tube into the initial slot (on the back of the skeleton) of the extension tendon (figure 5.28)
 2. Insert the other end into the final slot shown in figure 5.29
 3. Note: if the tube is too long, shorten it by cutting it with a utility knife.

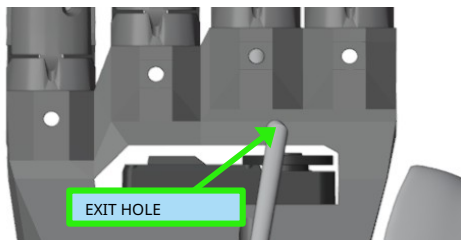


Figure 5.26: Inserting tube into the tendon exit hole (rear view)

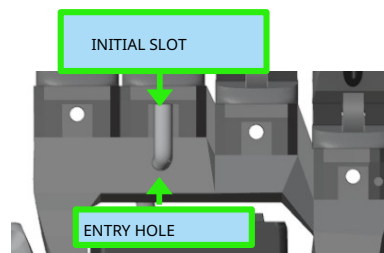


Figure 5.27: Inserting into the circular slot (front view)

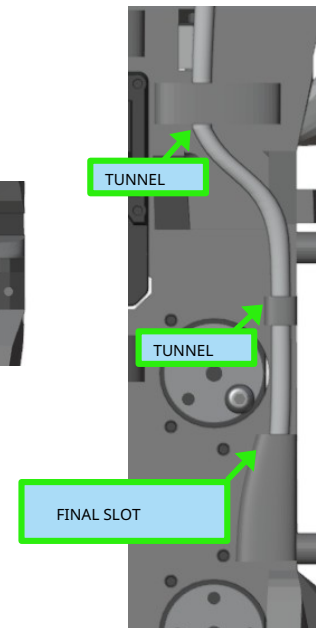


Figure 5.25 Inserting tube into the two passage tunnels and finally into the final slot (rear view)

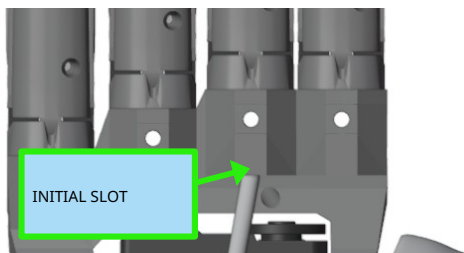


Figure 5.28 Inserting tube into the initial slot (rear view)

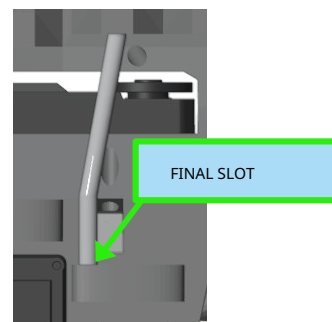


Figure 5.29 Inserting tube into the final slot (rear view)

5.9 PTFE tube installation Ring finger

For the RING finger only

1. Flexion tendon :
 1. Insert the PTFE tube into the exit hole (on the back of the skeleton) of the flexion tendon (figure 5.31), make it come out from the entry hole on the front of the skeleton and insert it into the initial slot figure 5.32)
 2. Take the remaining part (on the back of the skeleton) and thread it first into the passage tunnel and then into the final slot figure 5.30).
 3. Note: if the tube is too long, shorten it by cutting it with a cutter.
2. Extension tendon :
 1. Insert one end of the PTFE tube into the initial slot (on the back of the skeleton) of the extension tendon (figure 5.34)
 2. Insert the other end into the passage tunnel and then into the final slot figure 5.33).
 3. Note: if the tube is too long, shorten it by cutting it with a cutter.

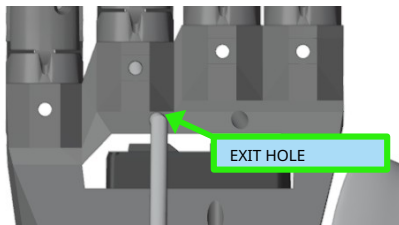


Figure 5.31: Tube insertion into the tendon exit hole (rear view)

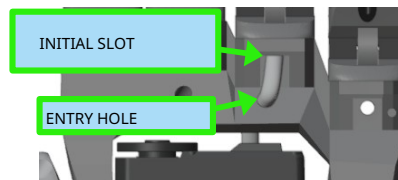


Figure 5.32: Insertion into the initial slot (front view)

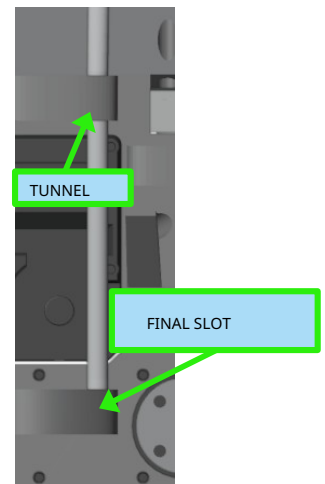


Figure 5.30:
Tube insertion into the passage tunnel and finally into the final slot (rear view)

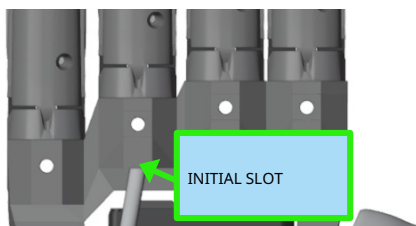


Figure 5.34: Tube insertion into the initial slot (rear view)

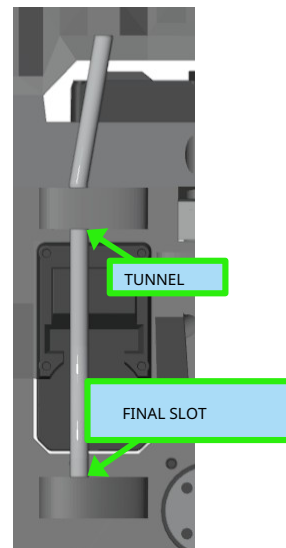


Figure 5.33:
Tube insertion into the passage tunnel and then into the final slot (rear view)

5.10 Little finger PTFE tube installation

For the little finger only

1. Flexion tendon:
 1. Insert the PTFE tube into the exit hole (on the back of the skeleton) of the flexion tendon (figure 5.37), pass it through the passage tunnel and make it come out from the entry hole on the front of the skeleton and insert it into the initial slot (figure 5.36).
 2. Take the remaining part (on the back of the skeleton) and insert it first into the passage tunnel and then into the final slot (figure 5.35).
 3. Note: if the tube is too long, shorten it by cutting it with a cutter.
2. Extension tendon:
 1. Insert one end of the PTFE tube into the initial slot (on the back of the skeleton) of the extension tendon (figure 5.39).
 2. Insert the other end into the passage tunnel and then into the final slot (figure 5.38).
 3. Note: if the tube is too long, shorten it by cutting it with a cutter.

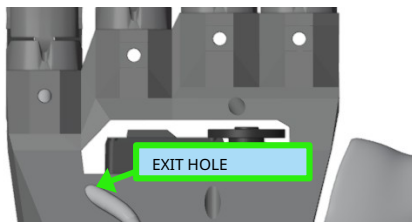


Figure 5.37 Tube insertion into the tendon exit hole (rear view)

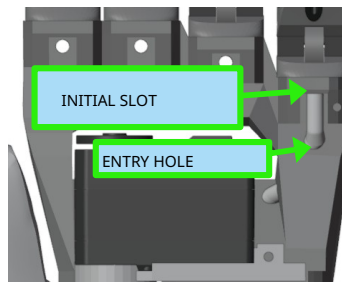


Figure 5.36 Insertion into the initial slot (front view)

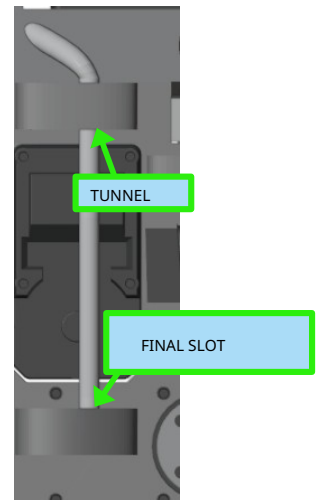


Figure 5.35
Tube insertion into the passage tunnel and finally into the final slot (rear view)

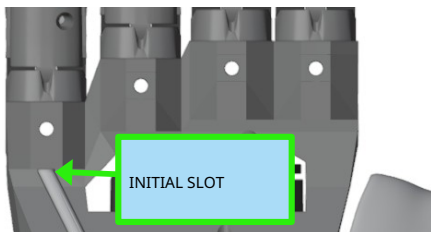


Figure 5.39 Tube insertion into the initial slot (rear view)

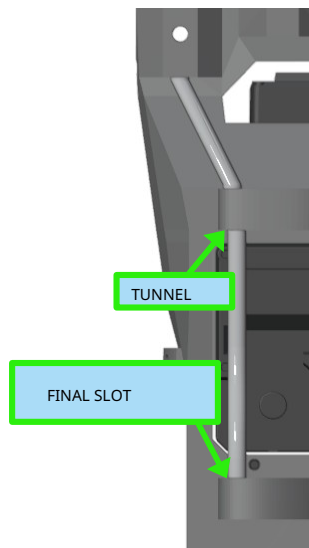


Figure 5.38
Tube insertion into the passage tunnel and then into the final slot (rear view)

5.11 Bowden tube installation check

Before proceeding, ensure that all bowden tubes are installed correctly as shown in the figure.

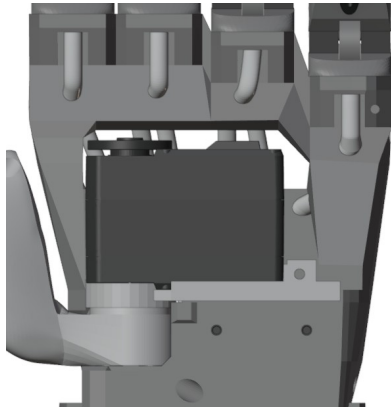


Figure 5.41 Front view with bowden tubes installed

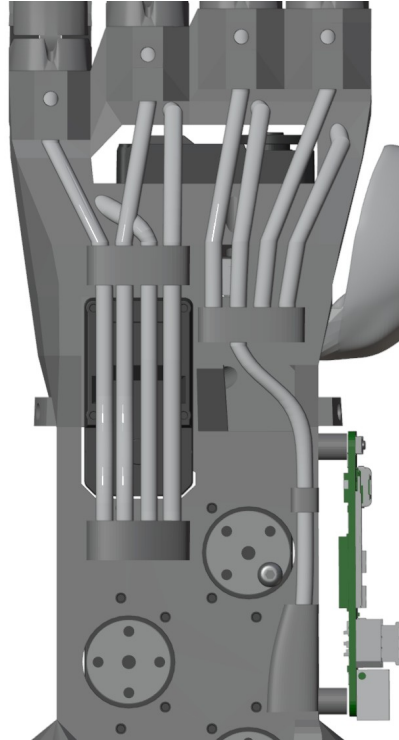


Figure 5.40: Rear view with bowden tubes installed

5.12 Inserting tendons into the bowden tubes (flexion end of flexion)

Procedure to be applied only to the finger tendons: index, middle, ring, and little finger.

For this operation, you need a nylon needle at least 20 cm long (40 cm wire).

For each finger:

1. On the front part of the skeleton, insert the two ends of the nylon needle into the entry hole in front of the finger slot (figure 5.43) and push them into the PTFE tube below, making them come out of the exit hole on the back part of the skeleton (figure 5.42)
2. Insert the flexion end of the tendon into the loop of the nylon needle that remained outside the entry hole (figure 5.43) and pull the ends of the nylon needle from the other end of the bowden tube, until the entire tendon comes out of the exit hole (figure 5.44).

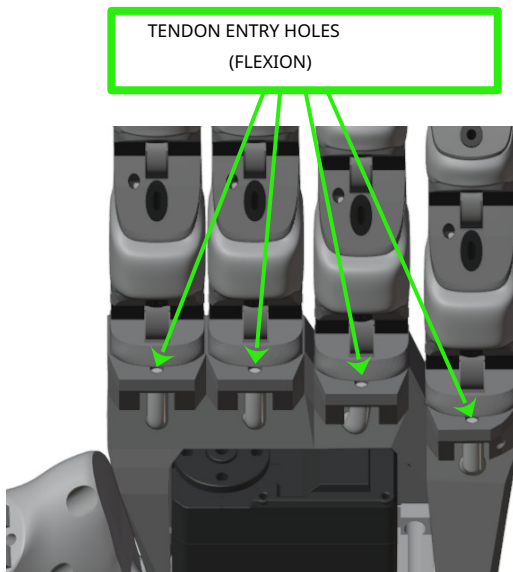


Figure 5.43: Tendon entry holes (flexion)

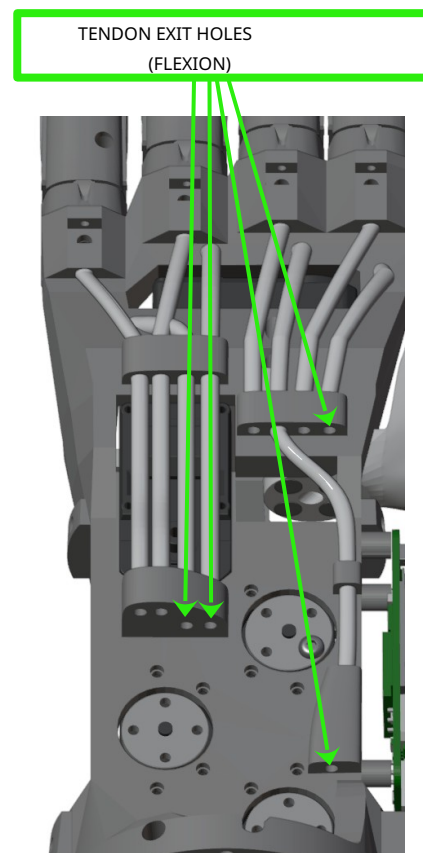


Figure 5.42: Tendon exit holes (flexion)

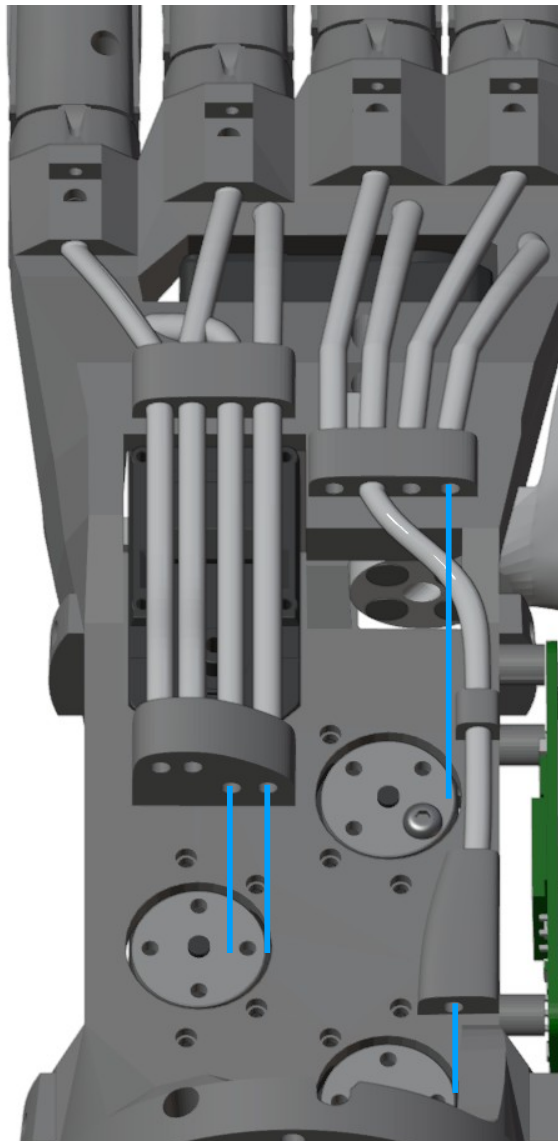


Figure 5.44: Flexion extremity of the tendons exiting the bowden tubes

5.13 Insertion of tendons into the bowden tubes (end of extension)

Procedure to be applied only to the finger tendons: index, middle, ring, and little finger.

For this operation, you need a nylon needle at least 20 cm long (40 cm wire).

For each finger:

3. On the back of the skeleton: insert the two ends of the nylon needle into the final hole of the finger and then into the underlying entry hole of the bowden tube (figure 5.46).
4. Push the nylon wires until they come out of the exit hole (figure 5.45).
5. Insert the extension end of the tendon into the loop of the nylon needle left outside the final hole of the finger (figure 5.46) and pull the ends of the nylon needle from the other end of the bowden tube, until the entire tendon comes out of the exit hole (figure 5.47)

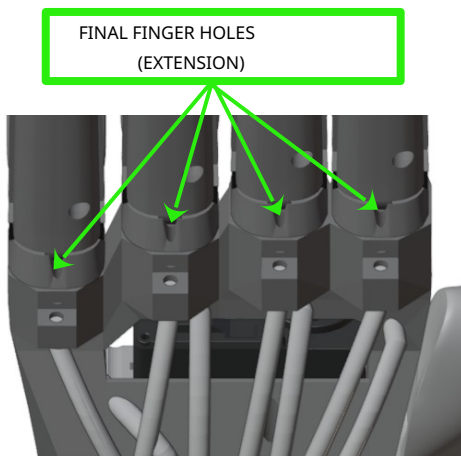


Figure 5.46: Tendon entry holes (extension)

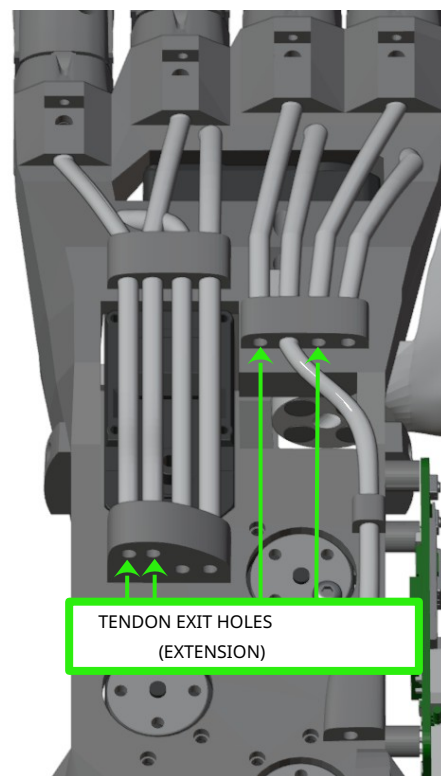


Figure 5.45: Tendon exit holes (extension)

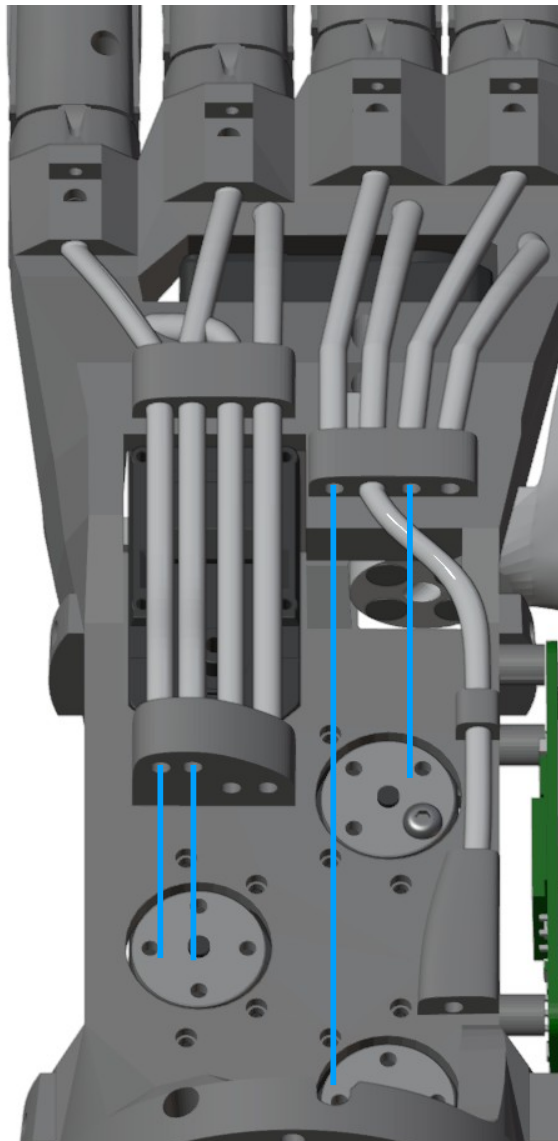


Figure 5.47: Extension extremity of the tendons exiting from the bowden tubes

5.14 Tendon installation on winders

5.14.1 Safety Information

!! DO NOT PULL THE TENDON WITH BARE HANDS !!

MANDATORY

- Use a vise to hold the part being worked on
- Use a manual assembly plier to pull the tendon

POSSIBLE DAMAGE:

- Severe finger cuts
- Bleeding



MANDATORY TO USE THE VISE



MANDATORY TO USE THE MANUAL ASSEMBLY PLIER



FINGER CUTTING HAZARD

5.14.2 Nylon needle technique

The tendons are thin and difficult to thread, which is why a special needle must be used.

This special needle is actually a nylon thread used as if it were a needle.

This method was chosen after multiple attempts, using various different tools, as it proved to be the fastest, most reliable, and safest.

The procedure for threading the tendon into a hole using a nylon thread (10 cm, 0.35 mm diameter) as a needle is shown below:



WARNING: Tendons can cut your fingers!

MANDATORY: Use the vise to hold the winder and the manual assembly pliers to pull the tendon.

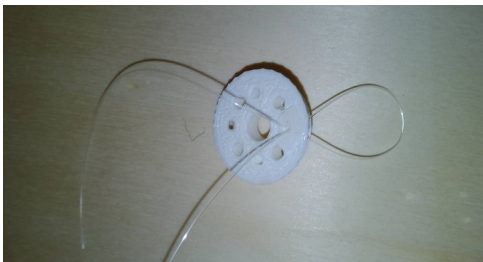


Figure 5.48: Thread the ends of the nylon thread, one at a time, into the hole



Figure 5.49: Thread the tendon into the nylon loop



Figure 5.50: Pull the two ends of the nylon loop from the other side of the hole with the manual assembly pliers



Figure 5.51: pull with the manual assembly pliers the shorter part of the tendon from the other side of the hole

NOTE: Hereafter we will refer to this sequence of operations simply by saying: "thread the tendon". It is implicitly assumed that it must be done using the nylon needle.

5.14.3 Tendon installation on winders (index, middle, ring and little finger)



WARNING: The tendons can cut your fingers!

MANDATORY: Use the vise to hold the skeleton and the manual assembly pliers to pull the tendon.

5.14.3.1 Winder assembly preparation

Each servo requires a flexion winder and an extension winder, which are mounted one on top of the other. See figure.

For each flexion winder (drawn in yellow in figure 5.52) :

1. Place the extension winder on top of it (drawn in blue in figure 5.52)
 1. Ensure that the flexion winder (in yellow in figure 5.52) has the smaller diameter at the bottom.
2. Secure the extension winder with 4 M3 10 mm button head bolts (figure 5.53); the screws will temporarily hold the assembly together by simple friction, without screwing (it is important that they remain joined until the end of the next phase 'Tendon installation on winders').



Figure 5.52:
Winder assembly



Figure 5.53: The
bolts keep the
assembly together

5.14.3.2 Tendon installation on the winders

Ensure that the winder assemblies are each correctly assembled (the screws temporarily hold the extension winder to the flexion winder by simple interlocking, without screwing)

Preparation:

1. Ensure that the servomotors are all connected, and that the winders are not installed.
2. Power the logic board using the power connector (ensure that the USB cable is disconnected) as shown in figure 5.54 .
3. Connect (PROVIDE BT SERIAL TOOL) via Bluetooth to the device (device name: YEAH-HAND-X-YYYYYY, where X is 'L' or 'R' depending on the model, and YYYYYY is the serial number)
4. send the 'INSTALL ' serial command via Bluetooth.
5. wait for the servomotors to complete their movement and proceed with the following paragraphs:
 1. Tendon installation in the winders (index, middle, ring and little finger)
 2. Tendon installation in the winders (thumb)

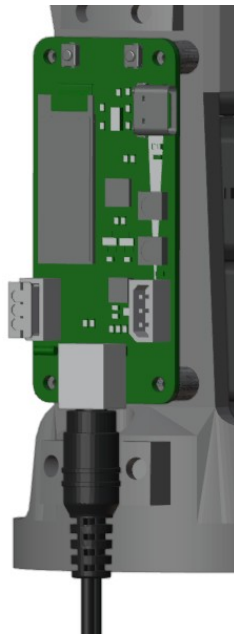


Figure 5.54:
Power connector
inserted

Tendon installation in the winders (index, middle, ring, and little finger):

1. For each servomotor:
 1. Identify the relative finger (or fingers in the case of ring and little finger) and the relative winder assembly (flexion winder and extension winder are currently held together by the inserted bolts, but not yet screwed into anything)
 2. Take the flexion end of the tendon (2 ends in the case of ring and little finger)
 3. Insert the end (or ends) into the bottom right hole of the flexion winder and pull it out from the corresponding hole of the extension winder (figure 5.56), ensure that the amount of tendon that has come out from the hole is long enough to prevent it from slipping out again.
 4. Ensure that the winder assembly does not disassemble during the operations.
 5. Take the extension end of the tendon (2 ends in the case of ring and little finger)
 6. Insert the end (or ends) into the top hole of the extension winder and pull it out from the bottom hole of the same winder (figure 5.55), ensure that the amount of tendon that has come out from the hole is long enough to prevent it from slipping out again.
 7. Position the winder assembly on the relative servomotor, with the arrow facing up figure 5.28)
 8. Screw the winder assembly to the flange of the relative servomotor with the M3 bolts
 9. Completion of the extension end of the tendon:
 1. Make the tendon perform a clockwise turn around the extension winder before it enters the upper hole (figure 5.60)
 2. Pull the tendon from the lower hole (until the finger is straight) and rotate it around the left bolt with a clockwise rotation around it (figure 5.59).
 3. While holding the tendon pulled with manual assembly pliers, tighten the bolt well with the wrench to lock the tendon (figure 5.58)
 10. Completion of the flexion end of the tendon:
 1. Pull the tendon until it is taut, without moving the position of the controlled finger, and rotate it around the right bolt with a clockwise rotation around it (figure 5.61).
 2. While holding the tendon pulled with manual assembly pliers, tighten the bolt well to lock the tendon (figure 5.62).

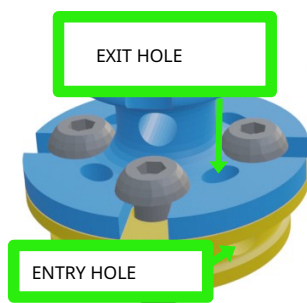


Figure 5.56: Insertion of flexion end



Figure 5.55:
Insertion of
extension
end

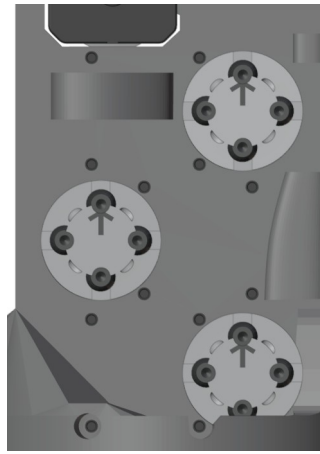


Figure 5.57: Assembled Winders mounted with arrow pointing up

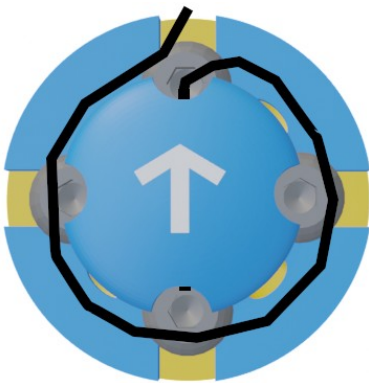


Figure 5.60: Pre-winding operation of the tendon extension end

Note: for ring and little finger, hook the additional tendon to the top bolt.

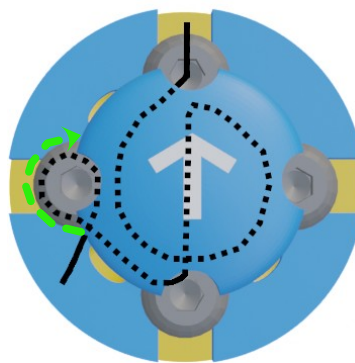


Figure 5.59: Extension end: Rotation around the bolt

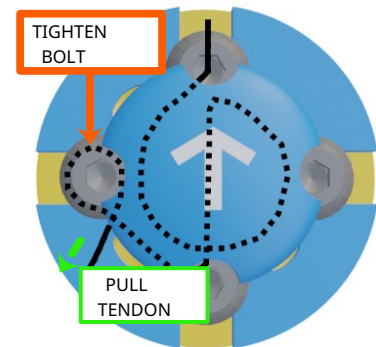


Figure 5.58: Extension end – Pulling and fastening

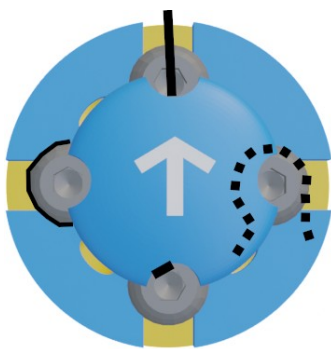


Figure 5.61: Flexion end: Rotation around the bolt

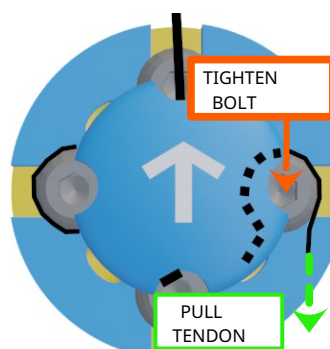


Figure 5.62: Flexion end: pulling and fastening

Note: for ring and little finger, hook the additional tendon to the bottom bolt

Tendon installation in the spoolers (thumb)

Note: the thumb uses a single spooler for both extension and flexion.

1. Only for the thumb servomotor
 1. Take the flexion end of the tendon
 2. Insert the end into the top right hole of the spooler and let it exit from the corresponding exit hole (figure 5.64), make sure that the amount of tendon coming out of the hole is long enough to prevent it from slipping out again.
 3. Take the extension end of the tendon
 4. Insert the end into the bottom left hole of the spooler and let it exit from the corresponding exit hole of the same spooler (figure 5.65), make sure that the amount of tendon coming out of the hole is long enough to prevent it from slipping out again.
5. Place the spooler on the relative servomotor and fix it with 4 M3 bolts (figure 5.66)
6. Pull the extension end of the tendon from the bottom hole (until the finger is straight) and fix it to the bottom bolt with a clockwise rotation around it (figure 5.67). While holding the tendon pulled with a manual assembly plier, tighten the bolt well to lock the tendon (figure 5.38). The figure 5.63 shows the completed operation.
7. Pull the flexion end of the tendon without moving the finger position and fix it to the top bolt with a clockwise rotation around it (figure 5.68). While holding the tendon pulled with a manual assembly plier, tighten the bolt well to lock the tendon (figure 5.68). The figure 5.69 shows the completed operation.

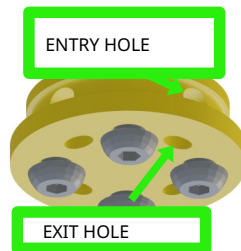


Figure 5.65:
Thumb Spooler :
Insertion
flexion end

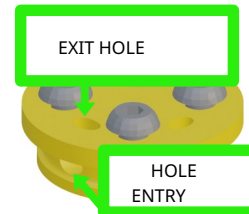


Figure 5.64:
Thumb Spooler
: Insertion of
extension end

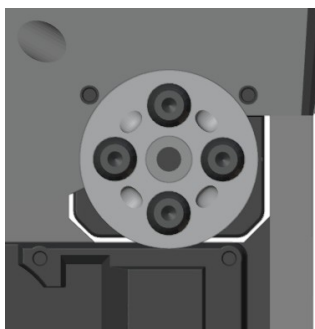


Figure 5.66:
Place spooler on
servomotor

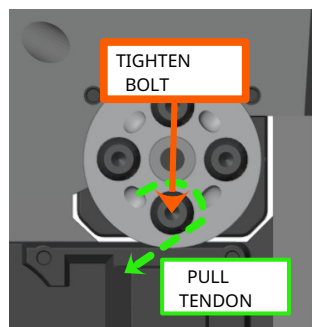


Figure 5.67: Rotation
around bolt and
fixing

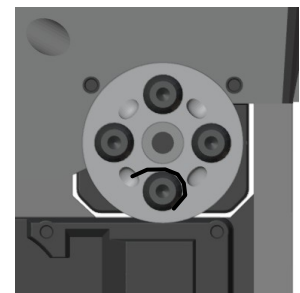


Figure 5.63: Tendon
fixed under the
bolt

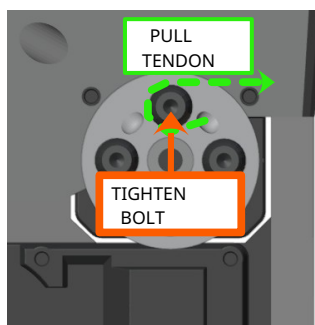


Figure 5.68: Rotation around bolt

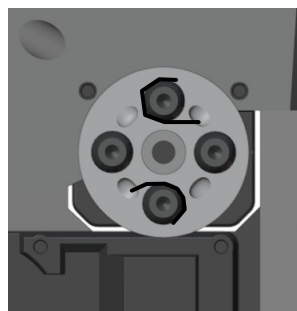


Figure 5.69: Tendon fixed under the bolt

5.14.4 Palm installation

Position the palm and secure it with 4 M2 screws 6 – 8mm long (see figure)

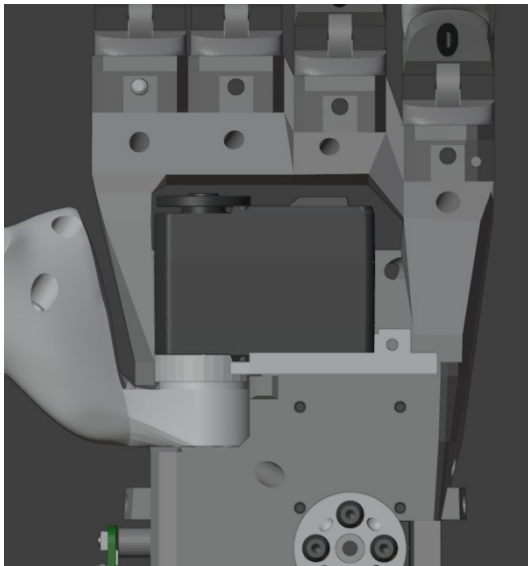


Figure 5.70: Skeleton - palm area



Figure 5.71: Skeleton - palm installed

5.14.5 Top cover installation

Position the thumb-side pins into the skeleton holes, snap them in, then rotating the cover proceed to snap the pins on the other side (see figures 5.73 and 5.72).

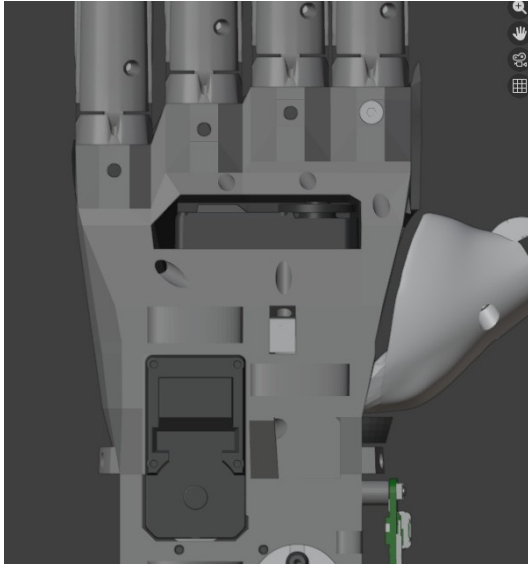


Figure 5.73: Skeleton - Top cover area

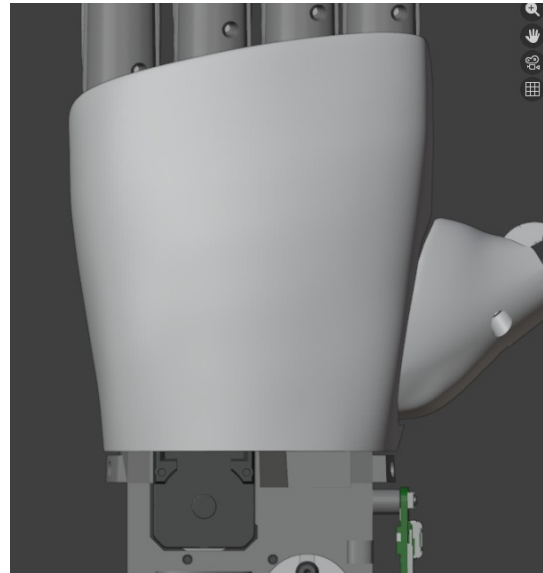


Figure 5.72: Skeleton - Top cover mounted

5.14.6 Installation of lower covers

1. Ensure that the logic board has already been mounted on the skeleton
2. Ensure that the logic board is connected to the servomotor at the base of the skeleton with the control cable
3. Position the pins on one side of the rear cover onto the interlocking holes on the corresponding side of the skeleton, snap them in, then rotating the cover proceed to snap the pins on the other side. See figures 5.74 and 5.75 .
4. Position the pins on one side of the front cover onto the interlocking holes on the corresponding side of the skeleton, snap them in, then rotating the cover proceed to snap the pins on the other side. See figures 5.76 and 5.77 .

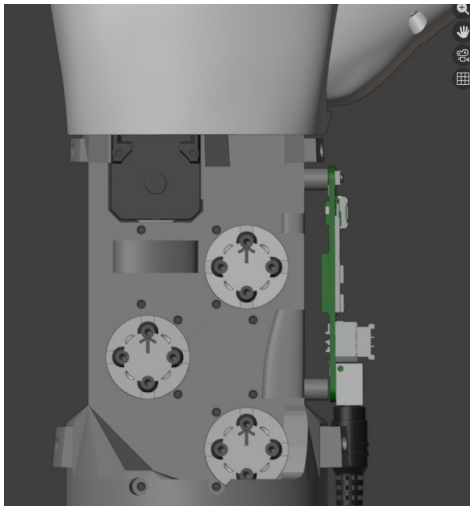


Figure 5.74: Skeleton - Rear lower cover area

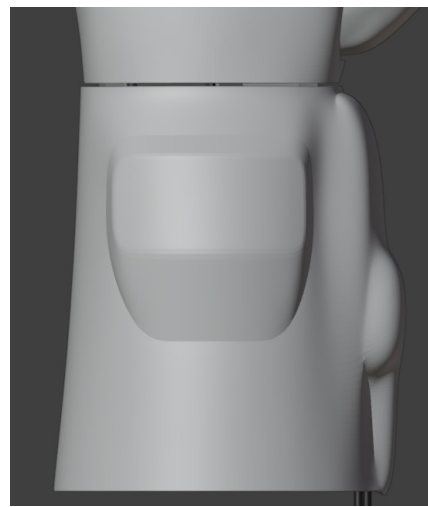


Figure 5.75: Skeleton - Rear lower cover installed

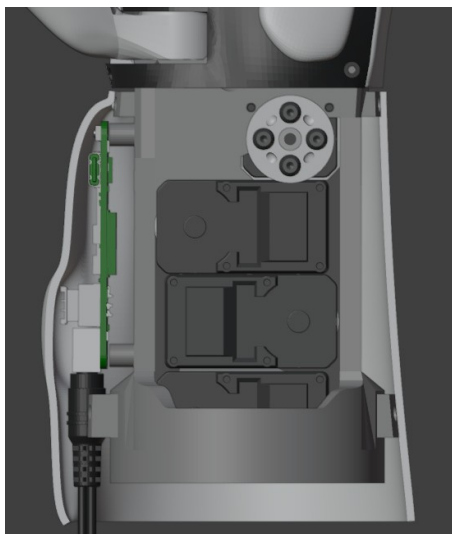


Figure 5.76: Skeleton - Front lower cover area

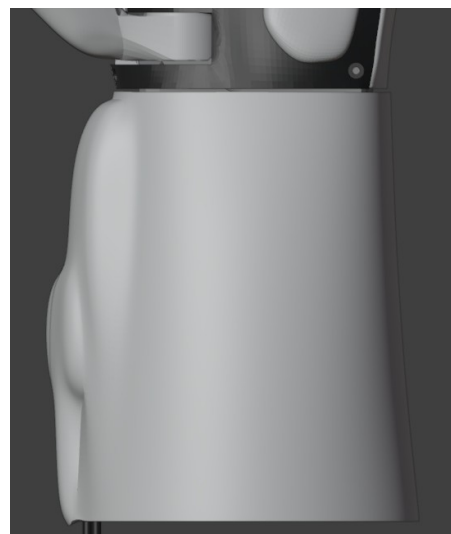


Figure 5.77: Skeleton - Front lower cover mounted

5.14.7 Finger cover installation

For each upper finger (index, middle, ring, little)

1. place the distal cover on the distal phalanx and press to snap it in
2. place the medial cover on the medial phalanx and press to snap it in
3. place the proximal cover on the proximal phalanx and press to snap it in

For the thumb:

1. place the distal cover on the distal phalanx and press to snap it in
2. place the proximal cover on the proximal phalanx and press to snap it in

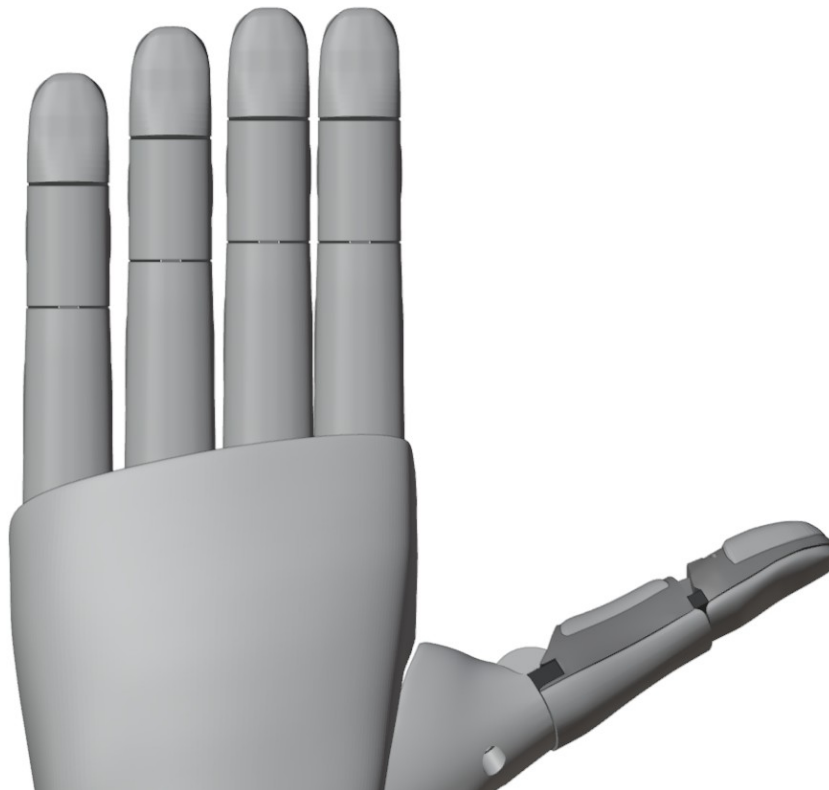
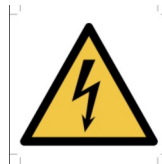


Figure 1: Covers installed on the fingers

6 Installation of the robotic hand on Robot Manipulator

1. Place the flange on the robot wrist, align with the alignment pin.
2. Screw in with 4 M6 countersunk bolts 3 cm long. TBD
3. Place the robotic hand on the flange according to the desired orientation.
4. Secure with 4 M3 countersunk bolts 3 cm long. TBD
5. Connect the robotic hand power cable connector to the robot connector wrist.

The provided logic board (WaveShare ESP32 Servo Driver Board) requires 12V DC. If it is necessary to power the robotic hand at 24V DC, you must install the voltage conversion board, which brings the voltage from 24V to 12V. In that case, the fuse to use is 2 A.



USE ONLY POWER SUPPLIES WITH THESE OUTPUT CHARACTERISTICS:

VOLTAGE: 12V DC
2A < CURRENT < 5A

Based on the available power supply, install the appropriate fuse (see table).

DC Power Supply	Fuse to install
12V 2A	2A
12V 4A	4A
12V 5A	5A
24V 2A + 12V 4A Conversion Board	4A
24V 1A + 12V 2A Conversion Board	2A

TODO INSERT 24/12 CONVERSION BOARD FIGURES

7 Original manufacturer settings of the equipment

The default settings of the robotic hand are these:

- Control mode
 - Grasp Control
 - MONKEY GRASP
- Finger position:
 - factor: 0% (maximum opening)
- Bluetooth device name:
 - YeahHand_XXX, where XXX is a unique code.

8 Operation



READ THE MANUAL BEFORE USING THE ROBOTIC HAND



READ THE INFORMATION ON TYPICAL OPERATION

8.1 Startup

To start the robotic hand, activate the power supply from the controls of the robotic arm on which it is installed, so that the robot wrist connector provides 12V and at least 162 W of power.

The robotic hand begins the tendon auto-calibration phase; once completed, the robotic hand will be in the maximum finger opening position, in grasping mode: monkey grasp.

Connect to the robotic hand from the robot control system, via Bluetooth character serial connection (device name: YEAH-HAND-L_XXXXX or YEAH-HAND-R_XXXXX, where XXXXX is the serial number of the robotic hand).

The serial commands are in the format `#CMD##\r\n`, where CMD is the command string.

Examples:

- Command: `#MONKEY##\r\n`
 - The command string is " MONKEY "
- Command: `#I%dM%dR%dT%dTR%d ##\r\n`
 - The command string is " I%dM%dR%dT%dTR%d "

8.2 Control mode selection

Control mode	Full command
Monkey grasp	#MONKEY##
Power grasp	#POWER##
PowerSmall grasp	#POWERSMALL##
Pinch grasp	#PINCH##
Relax	#RELAX##
RAW	#RAW##

8.3 Finger movement

8.3.1 Grasp Control mode

If a mode among MONKEY, POWER, POWERSMALL, PINCH has been selected, the robotic hand can be controlled in this way:

- Command: `##d##\r\n`
- Meaning: %d expresses the integer value [0, 100] expressing the finger closing percentage (each type of selected grasp applies the closing to all fingers differently from the other grasps)

Examples:

EXAMPLE	FULL COMMAND
Maximum opening	<code>#0##\r\n</code>
Maximum closing	<code>#100##\r\n</code>
50% closing	<code>#50##\r\n</code>
75% closing	<code>#75##\r\n</code>

8.3.2 RAW Mode

If a 'RAW' mode has been selected, the robotic hand can be controlled in this way:

- Command: #I%dM%dR%dT%dTR%d##\r\n
- Meaning:
 - # : initial character of the data packet
 - I%d : the value %d contains an integer with the closure value [0, 100] index flexion
 - M%d : the value %d contains an integer with the closure value [0, 100] middle flexion
 - R%d : the value %d contains an integer with the closure value [0, 100] ring flexion
 - T%d : the value %d contains an integer with the closure value [0, 100] thumb flexion
 - TR%d : the value %d contains an integer with the closure value [0, 100] thumb rotation
 - ##\r\n : final character sequence of the data packet

Examples:

EXAMPLE	FULL COMMAND
Index Flexion: 30% Middle Flexion: 50% Ring and little finger flexion: 10% Thumb Flexion: 70% Thumb Rotation: 65%	#I30M50R10T70TR65##\r\n
Index Flexion: 10% Middle Flexion: 20% Ring and little finger flexion: 30% Thumb Flexion: 40% Thumb Rotation: 50%	#I10M20R30T40TR50##\r\n
Flexions : maximum opening Thumb rotation: maximum opening (open hand)	#I0M0R0T0TR0##\r\n
Flexions : maximum closure Thumb rotation: maximum closure (thumb in opposition)	#I0M0R0T0TR100##\r\n

9 Product or capacity change

Before replacing an object picked up by the robotic hand, ensure that its characteristics fall within the limits set out in the chapter [Technical characteristics of the machine](#) in the section "Object properties" .

10 Inspections, tests and maintenance

10.1 Expected component lifespan

Note: A cycle is a complete closing of the finger from the 0% position (fully open) to the 100% position (fully closed, equal to 300° of servomotor rotation) and subsequent reopening to the 0% position.

COMPONENT	LIFESPAN IN CYCLES
Skeleton	200000
TRS Platform	200000
Thumb base	20000
Fingers	20000
Winders	20000
Covers	200000
Tendons and bowden tubes	200000
Servomotors	20000
Logic board	200000

10.2 Nature - frequency and inspection criteria

COMPONENT	INSPECTION INTERVAL IN CYCLES
Skeleton	50000
TRS Platform	50000
Thumb base	5000
Fingers	5000
Winders	5000
Covers	50000
Tendons and bowden tubes	50000
Servomotors	5000
Logic board	50000

10.3 Spare parts specifications

All spare parts are specified in the following table:

DESCRIPTION	QUANTITY IN A PRODUCT	MANUFACTURER	MODEL
skeleton	1	VITTORIO- LUMARE	H-SKELETON-X
index with silicone pads	1	VITTORIO- LUMARE	H-FING-IF-X
middle with silicone pads	1	VITTORIO- LUMARE	H-FING-MF-X
ring with silicone pads	1	VITTORIO- LUMARE	H-FING-RF-X
little with silicone pads	1	VITTORIO- LUMARE	H-FING-LF-X
thumb with silicone pads	1	VITTORIO- LUMARE	H-FING-TF-X
palm with silicone pads	1	VITTORIO- LUMARE	H-PALM-X
thumb base	1	VITTORIO- LUMARE	H-FING-TB-X
bowden tubes	1	VITTORIO- LUMARE	H-PTFE-TUBING-X
tendon spool	4	VITTORIO- LUMARE	H-SPOOL-X
front cover	1	VITTORIO- LUMARE	R-COV-F-X
rear cover	1	VITTORIO- LUMARE	R-COV-B-X
index cover assembly	1	VITTORIO- LUMARE	H-FING-COV-IF-X
middle cover assembly	1	VITTORIO- LUMARE	H-FING-COV-MF-X
ring cover assembly	1	VITTORIO- LUMARE	H-FING-COV-RF-X
little cover assembly	1	VITTORIO- LUMARE	H-FING-COV-LF-X
thumb cover assembly	1	VITTORIO- LUMARE	H-FING-COV-TF-X
flange for UR3	1	VITTORIO- LUMARE	R-FLANGE-UR3-X
servomotors and cables	5	WAVESHARE	ST3215-HS
logic board	1	WAVESHARE	Servo Driver with ESP32
polyethylene tendons (skein)	1	HERCULES	90.7 kg, 0.75mm, 8 strands
M3 bolts, countersunk head, 20 mm long	TBD	N.A.	N.A.
M3 bolts, countersunk head, 16 mm long	TBD	N.A.	N.A.
M3 bolts, countersunk head, 12 mm long	TBD	N.A.	N.A.
M3 bolts, button head, 10 mm long	TBD	N.A.	N.A.
M6 bolts, hex head, 35 mm long	TBD	N.A.	N.A.
M3 bolts, countersunk head, 20 mm long	TBD	N.A.	N.A.
M2 bolts, socket head cap, 5 mm long	TBD	N.A.	N.A.
M2 self-tapping screws 6mm long	TBD	N.A.	N.A.

10.4 Equipment

10.4.1 Assembly and maintenance equipment

These are the tools and equipment required for assembly

DESCRIPTION	QUANTITY	MANUFACTURER	MODEL
Phillips #1 long screwdriver for electronics 6.69"	1	WIHA	261110
Precision pliers 4.72"	1	ADAFRUIT INDUSTRIES LLC	422
Hex screwdriver 2mm 5.71"	1	WIHA	26320
Hex screwdriver 1.5mm 7.05"	1	WIHA	36715
Small manual assembly pliers	1	OLIMEX LTD	CHN-PLI-LC

10.5 Energy isolation procedures

To disconnect the electrical power source that powers the robotic hand, simply unplug the power plug from the side connector (figure 10.1).

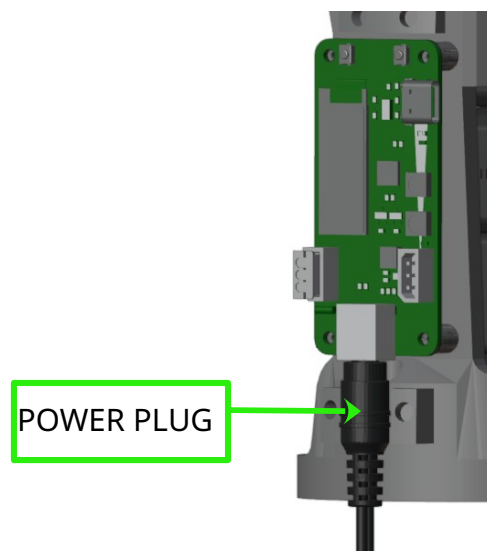


Figure 10.1:
Connector
with power
plug inserted

10.6 Extraordinary Maintenance

These operations are only for cases where a component requires replacement.

10.6.1 Finger replacement

Perform in sequence:

1. Tendon uninstallation
2. Finger uninstallation
3. Finger installation
4. Tendon installation
5. Tendon adjustment

10.6.2 Finger uninstallation



USE THE VISE

1. Unscrew all finger fixing bolts
2. Remove all fingers from their slots

10.6.3 Finger installation



USE THE VISE

See chapter #[5.1.4.Upper finger installation | outline](#)

See chapter #[5.1.5.Thumb installation | outline](#)

10.6.4 Tendon uninstallation



1. Cut all tendons with scissors
2. Unscrew the tendon fixing bolts from all the winders using the 2.5mm hex key provided
3. Remove all tendons using the provided manual assembly pliers

10.6.5 Tendon installation



See chapter [#5.1.6.Tendon installation|outline](#)

10.6.6 Motor replacement

Perform in sequence:

1. Motor uninstallation
2. Motor installation

10.6.7 Motor Uninstallation

1. Remove each servomotor by unscrewing the 4 fixing screws, following this sequence:
 1. ring-little finger flexion servomotor
 2. index finger flexion servomotor
 3. middle finger flexion servomotor
 4. thumb flexion servomotor
 5. thumb rotation servomotor

10.6.8 Motor Installation

See chapter [#5.1.1.Servomotor assembly|outline](#)

10.6.9 Cover replacement

Perform in sequence:

1. Cover uninstallation
2. Cover installation

10.6.10 Cover uninstallation

1. Remove the snap-fit front cover (palm side)
2. Remove the snap-fit rear cover (back side)

10.6.11 Cover installation

See chapter [Cover installation](#)

10.6.12 Skeleton replacement

To replace the skeleton, it is necessary to disassemble all components and reassemble the robotic hand from the beginning.

Perform in sequence:

1. Cover uninstallation
2. Board uninstallation
3. Tendon uninstallation
4. Bowden tube uninstallation
5. Winder uninstallation
6. Finger uninstallation
7. Servomotor uninstallation including TRS Platform
8. SKELETON REPLACEMENT
9. Servomotor installation including TRS Platform
10. Finger installation
11. Winder installation
12. Bowden tube installation
13. Tendon installation
14. Board installation
15. Cover installation

10.7 Tendon adjustment

For each finger requiring adjustment:

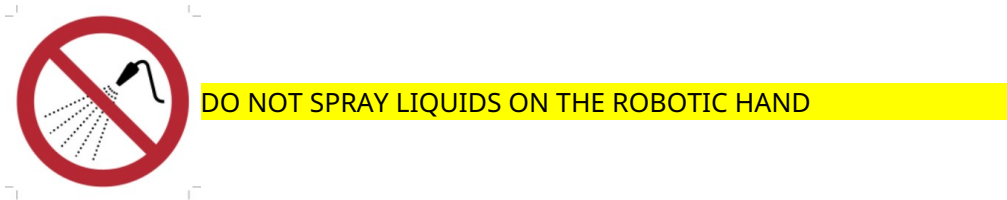
1. Secure the robotic hand in the vise
2. Connect the servomotor in the SSI slot to the electronic board.
3. Power the electronic board with a 12V DC power supply capable of delivering up to 14A, with a unipolar connector with a central positive hole
4. Connect via Bluetooth to the logic board.
5. Send the selection serial command ' Tendon Installation Mode' .
6. Wait for the servomotors to complete their movement and proceed to the next point
7. For each finger requiring adjustment
 1. Unscrew the extension tendon bolt by 2 turns using the 2.5mm hex key
 2. Unscrew the flexion tendon bolt by 2 turns using the 2.5mm hex key
 3. Ensure the finger is fully extended, otherwise extend it manually
 4. Using the manual assembly pliers with one hand, grab the extension tendon and pull it until it is taut
 5. While keeping the tendon pulled with the manual assembly pliers, with the other hand tighten the fixing bolt until it is well secured TBD
 6. Release the tendon with the manual assembly pliers
 7. Ensure the finger is fully extended under the effect of the tendon pull
 8. Using the manual assembly pliers with one hand, grab the flexion tendon and pull it until it is taut

9. Keeping the tendon pulled with the manual assembly pliers, with the other hand tighten the fixing bolt until it is tight TBD
10. Release the tendon with the manual assembly pliers

11 Cleaning and disinfection

The external surfaces of the robotic hand can be cleaned with a cloth slightly dampened with alcohol.

The external surfaces of the robotic hand can be cleaned with a cloth slightly dampened with disinfectant.



12 Fault detection / troubleshooting

ANOMALY	CAUSE
• Limp/oscillating fingers even if not moved by the motor • Fingers closing late • Fingers not closing completely	• tendons not tensioned (robotic hand used beyond load limits)
• Fingers not opening completely • Fingers not closing completely	• tendon calibration failed
• Fingers not moving	• damaged tendons • detached tendon • damaged motors • power supply interruption • damaged winders
• The motors stop exerting force after a certain time	• motor overheating
• Cannot establish a Bluetooth connection with the hand robotics	• Distance too great from the controlling device or interference or obstacles
• Fingers not gripping the object	• Worn pads
• Damaged or worn parts	• Breakage or wear and tear • Breakage or improper use

CAUSE	SOLUTION
• Tendons not tensioned	• remove the cover and perform tendon registration
• Damaged tendons	• replace the tendons
• Tendon calibration failed	• Restart the robotic hand to repeat the auto-calibration. • Turn off the robotic hand, remove the covers and check that the tendons have not become stuck together or with other parts of the robotic hand
• Detached tendons	• Turn off the robotic hand, remove the covers and reattach or replace the tendon, then perform tendon registration
• Motor overheating	• Ensure the robotic hand is used within the correct operating range
• Distance too great from the controlling device or interference or obstacles	• Move the controlling device closer within the bluetooth range (<9m) • Remove obstacles between the controlling device and the robotic hand • Eliminate any interference
• Worn pads	• Replace the parts containing the damaged pads
• Power supply interruption	• Check that the power supply is on and properly connected • Check that the power cables are not damaged
• Breakage or wear and tear • Breakage or improper use	• Replace damaged parts with new or refurbished parts from the manufacturer

13 Disassembly, deactivation and scrapping

13.1 Uninstall the robotic hand

Before uninstalling the robotic hand, deactivate it by disconnecting the power supply cable.

Perform in sequence to disassemble the robotic hand from the robotic manipulator:

- unscrew the 4 M3 bolts holding the flange attached to the robotic hand
- detach the robotic hand from the flange
- unscrew the 4 M6 bolts holding the flange attached to the manipulator wrist
- detach the flange from the manipulator wrist

13.2 Deactivate the robotic hand

To deactivate the robotic hand, detach the power cable.

13.3 Disassembly

Before disassembling the robotic hand, deactivate it and detach it from the machine on which it is installed.

To disassemble the robotic hand, perform in sequence:

- remove the robotic hand covers
- remove the finger covers
- remove the palm
- remove the bolts from the winders
- remove the winders
- remove the tendons with the manual assembly pliers, if necessary cut them with scissors
- remove the fingers: index, middle, ring, little
- remove the flexible part of the thumb
- remove the base of the thumb
- remove the cable ties
- remove the tendon-barrier
- remove the motors

13.4 Scrapping

The product, when it reaches the end of its life, must not be disposed of with undifferentiated municipal waste.

If delivered intact (not disassembled), it must be treated as Waste Electrical and Electronic Equipment (WEEE) and delivered to municipal collection centers, authorized collection points, or other WEEE collection systems provided locally.

Alternatively, the user can proceed with the disassembly of the product and deliver separately the different types of components according to the methods indicated in the following paragraphs and in compliance with national and local waste regulations.

13.5 Electronic components

Electronic components (servomotors, cables, control board) must be disposed of according to WEEE disposal regulations.



13.6 Bolts, nuts and washers

Bolts, nuts, washers, and other small metal parts, if separated from other components, must be delivered to separate collection systems dedicated to metals or multi-material (where provided), according to the methods established by the local waste management system.

The user must comply with the specific regulations in force in their country and the indications of the local waste manager.

13.7 Silicone pads

Parts containing silicone (flexible fingers, palm, and similar components) consist of already catalyzed silicone and do not contain substances known to be hazardous to the end user.

In the absence of dedicated collection streams, such components can be disposed of as undifferentiated municipal waste, in compliance with the specific regulations in force in their country and the indications of the local waste manager.

13.8 Rigid plastic parts

Rigid plastic parts (skeleton, thumb base, Bowden tubes, winders, covers, flange and similar components), if separated from electronic and metal parts, must be disposed of in separate collection circuits dedicated to plastics or multi-material, or at ecological islands/collection centers, according to the local waste management system. The user must comply with national regulations and the instructions of the local operator regarding the collection and recycling of plastic waste.

13.9 Tendons

The tendons are made of high-density polyethylene (HDPE), a high-density plastic that must not be dispersed in the environment.

These components must be disposed of through separate collection systems for plastic or at ecological islands/collection centers that accept non-hazardous plastic waste, according to the local waste management system. The user must comply with current regulations and the instructions of the local operator to ensure the correct recovery or disposal of this material.



14 Documents and drawings

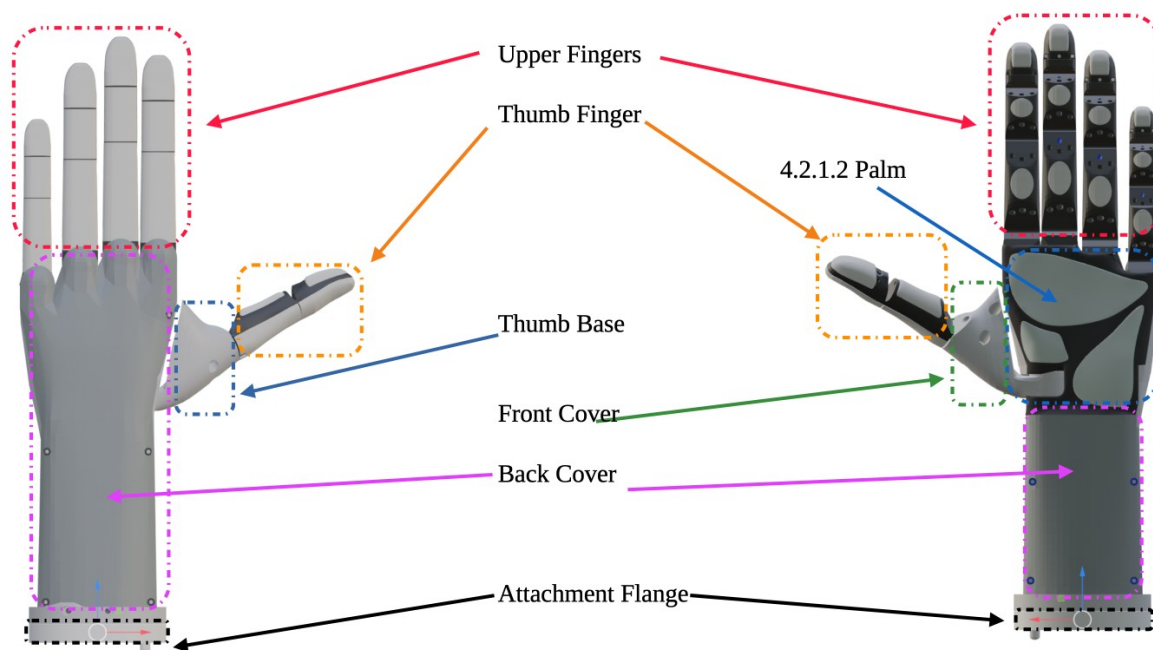


Figure 14.1: General drawing - left hand - external view

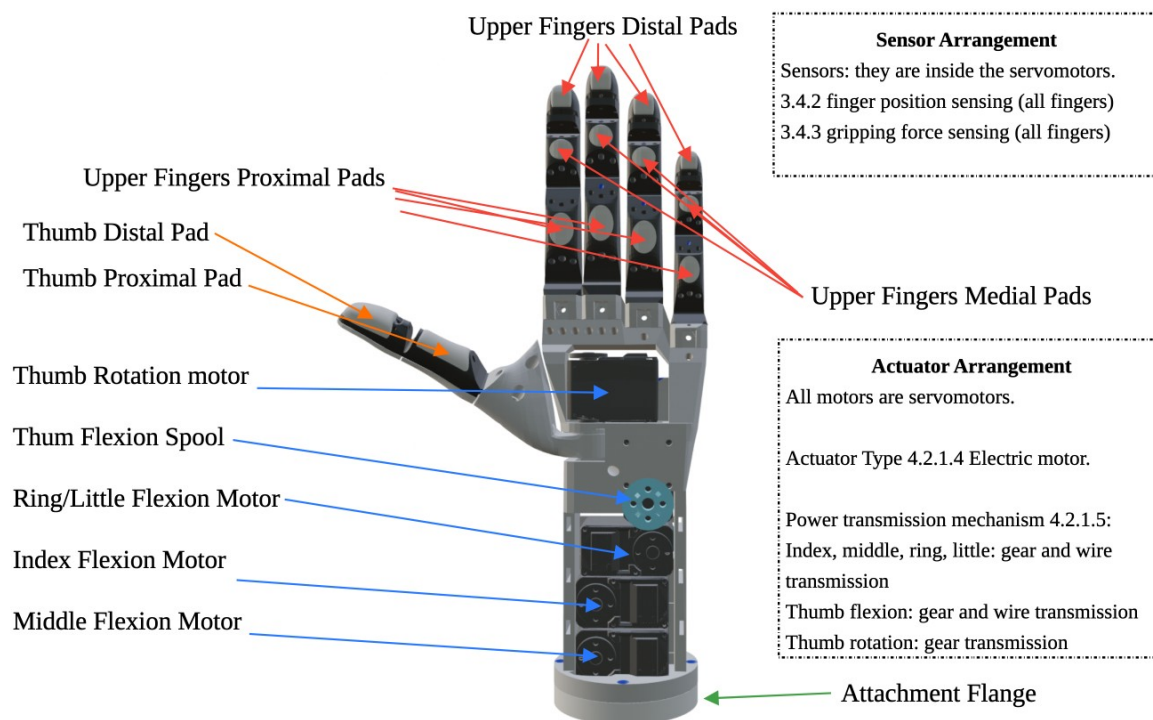


Figure 14.2: Detailed drawing - left hand - Internal view

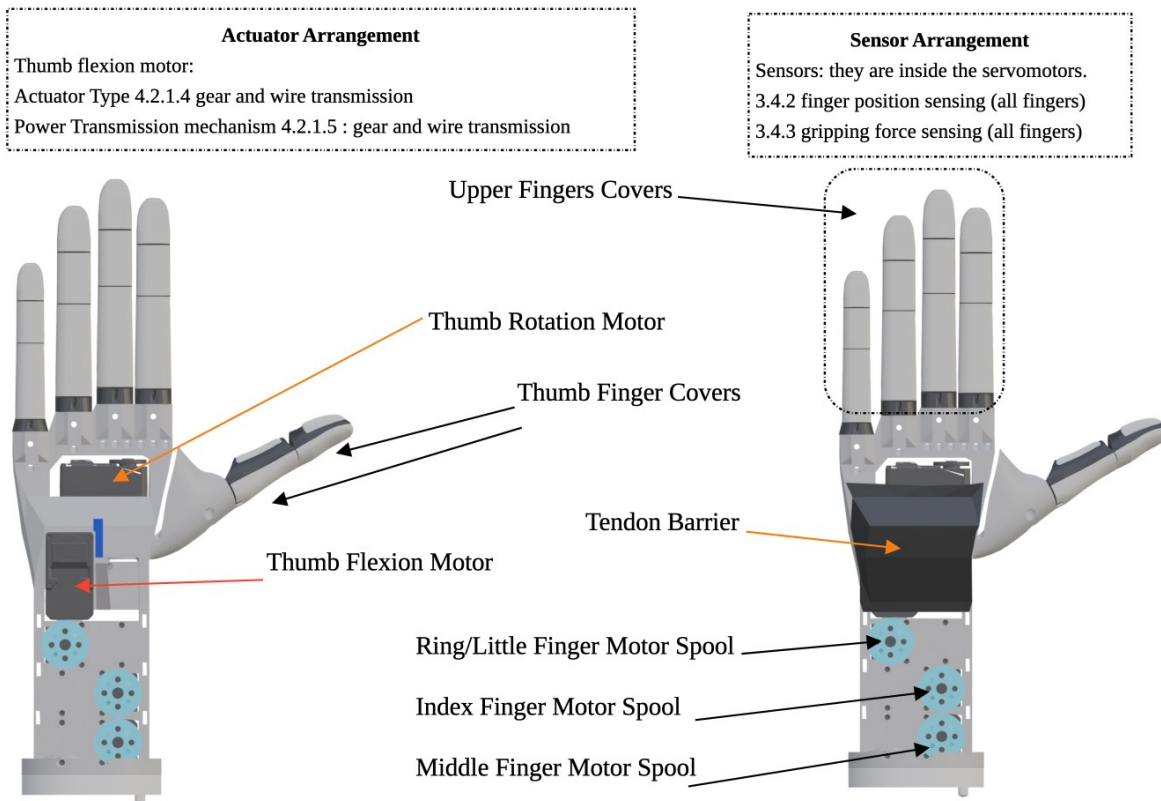


Figure 14.3: Left hand - Actuator and sensor arrangement

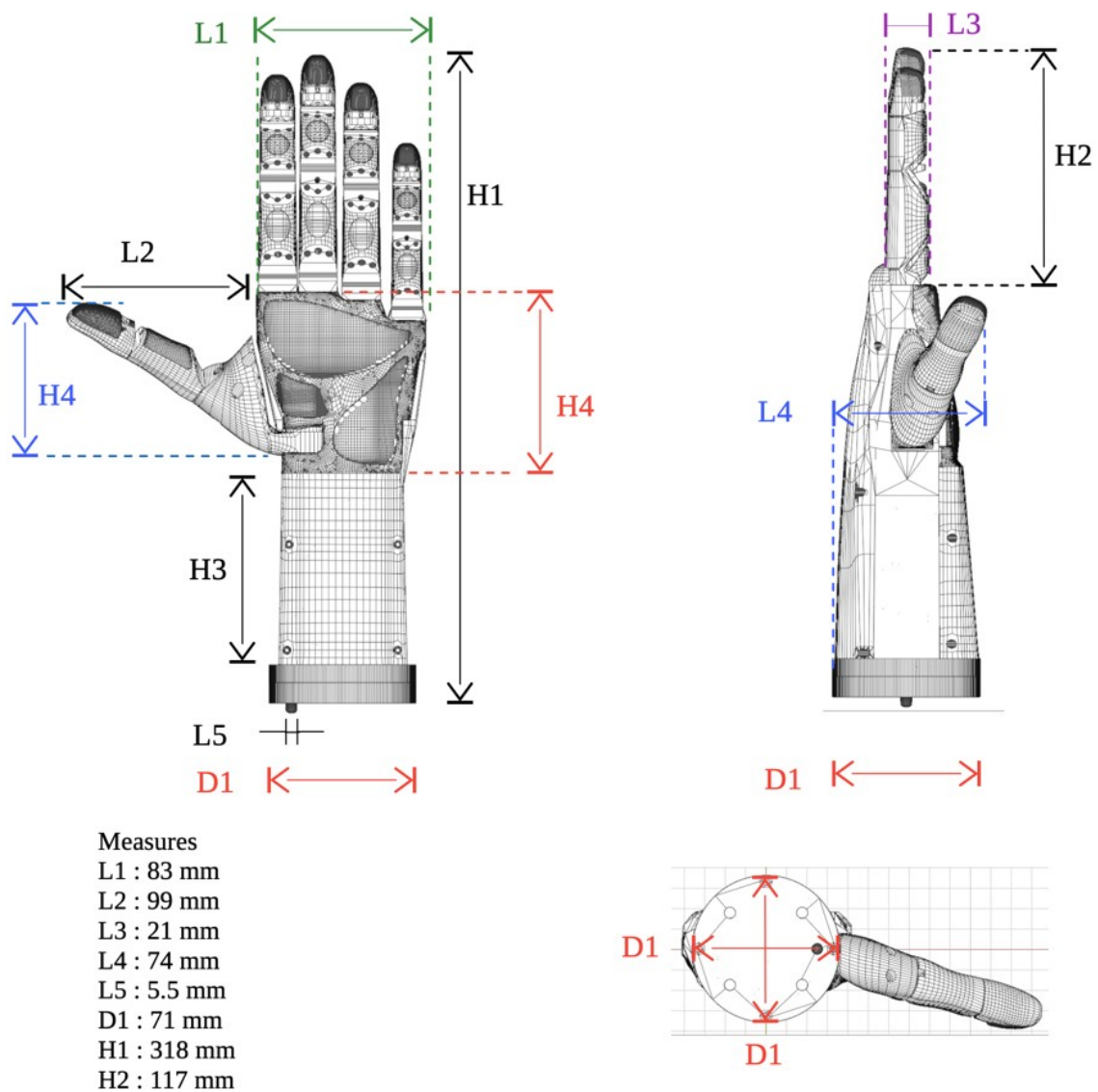
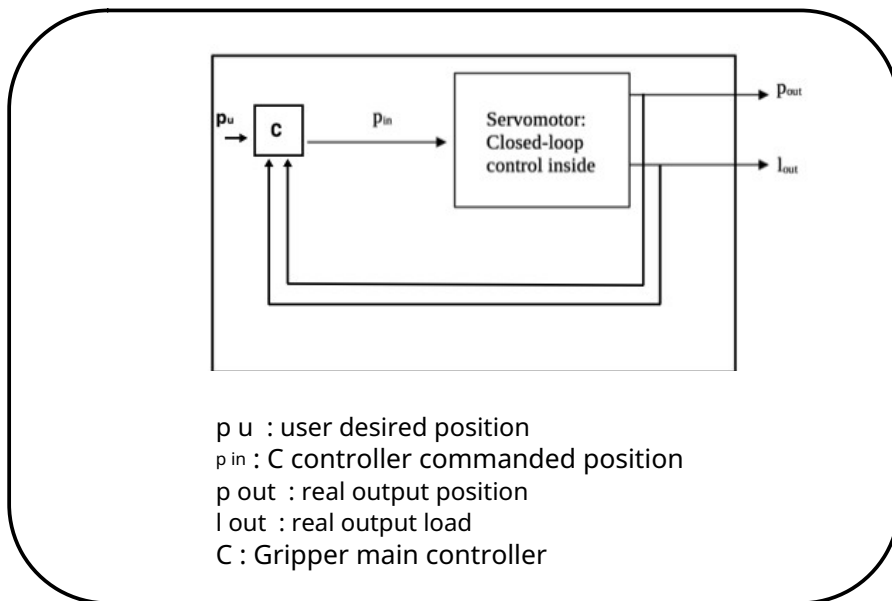


Figure 14.4: Left hand - Measurements



Drawing 1: Block diagram - Finger flexion/extension control system

14.1 Components

Components:

- skeleton
- fingers
 - index with silicone pads
 - middle with silicone pads
 - ring with silicone pads
 - pinky with silicone pads
 - thumb with silicone pads
- servomotors and cables : 5 x WaveShare ST3215HS
- logic board : Waveshare ESP32 Servo Driver Board
- tendon winders
- polyethylene tendons: Hercules 8 strand polyethylene brand, diameter 0.8 mm
- hardware
 - M3 countersunk head bolts
 - length 20mm
 - length 16mm
 - M3 button head bolts: length 10mm
 - M3 hex socket countersunk head bolts
 - length 12mm
 - M6 bolts
 - length 3.5cm
 - M2 self-tapping screws
 - length 6mm
- thumb base
- TRS platform (Thumb Rotation Servo platform)
- palm with silicone pads
- covers
 - upper rear cover
 - lower front cover
 - lower rear cover
 - finger phalanx covers
 - mechanical connection flanges
 - flange for UR3 manipulator robot (for the final joint on the wrist)

14.2Part nomenclature

Note: holes are indicated in red.

14.2.1 Thumb servomotor hole nomenclature

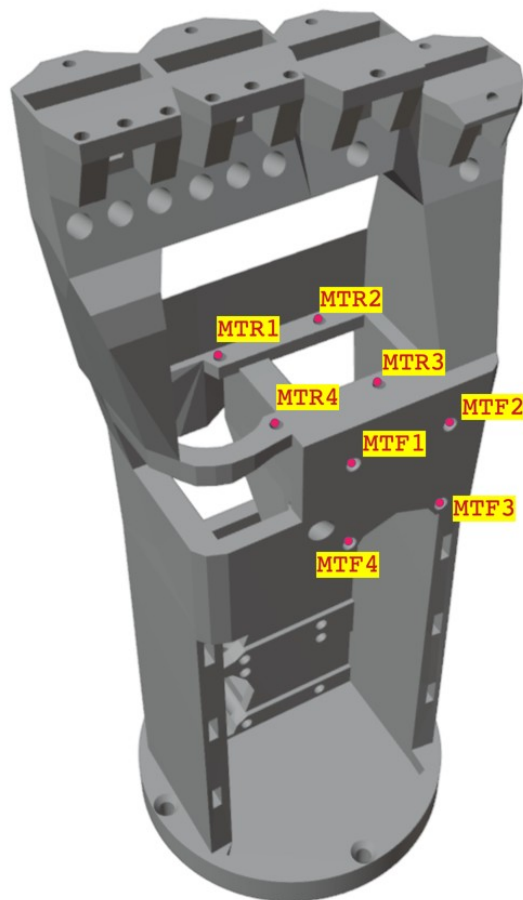


Figure 14.5: Left hand skeleton -
anterior view - servomotor holes

14.2.2 Nomenclature of upper finger servomotor holes

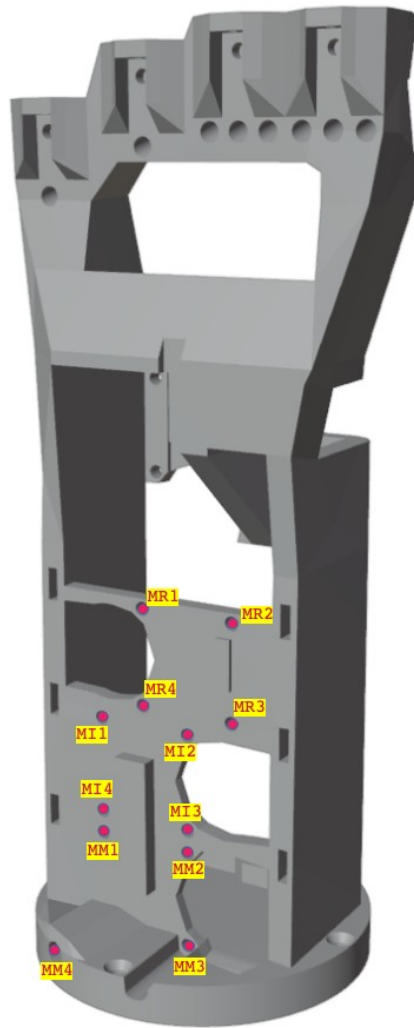


Figure 14.6: Left hand skeleton - rear view - servomotor holes

14.2.3 Servomotor slot nomenclature

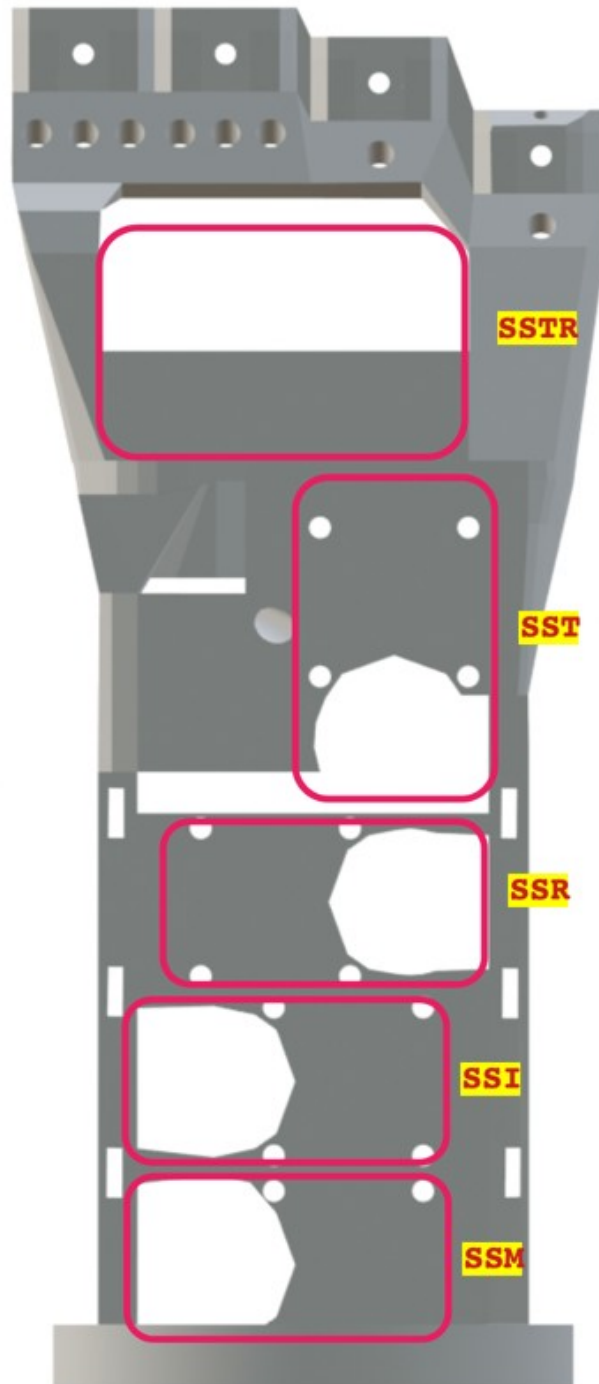


Figure 14.7: Left hand skeleton - anterior view - motor slots

14.2.4 Servomotor flange hole nomenclature

Figure 14.8: Left hand - front view - tendon barrier holes



Figure 14.9: Waveshare ST3215 ST3215/HS servomotor - holes

14.2.5 Upper finger slot and hole nomenclature

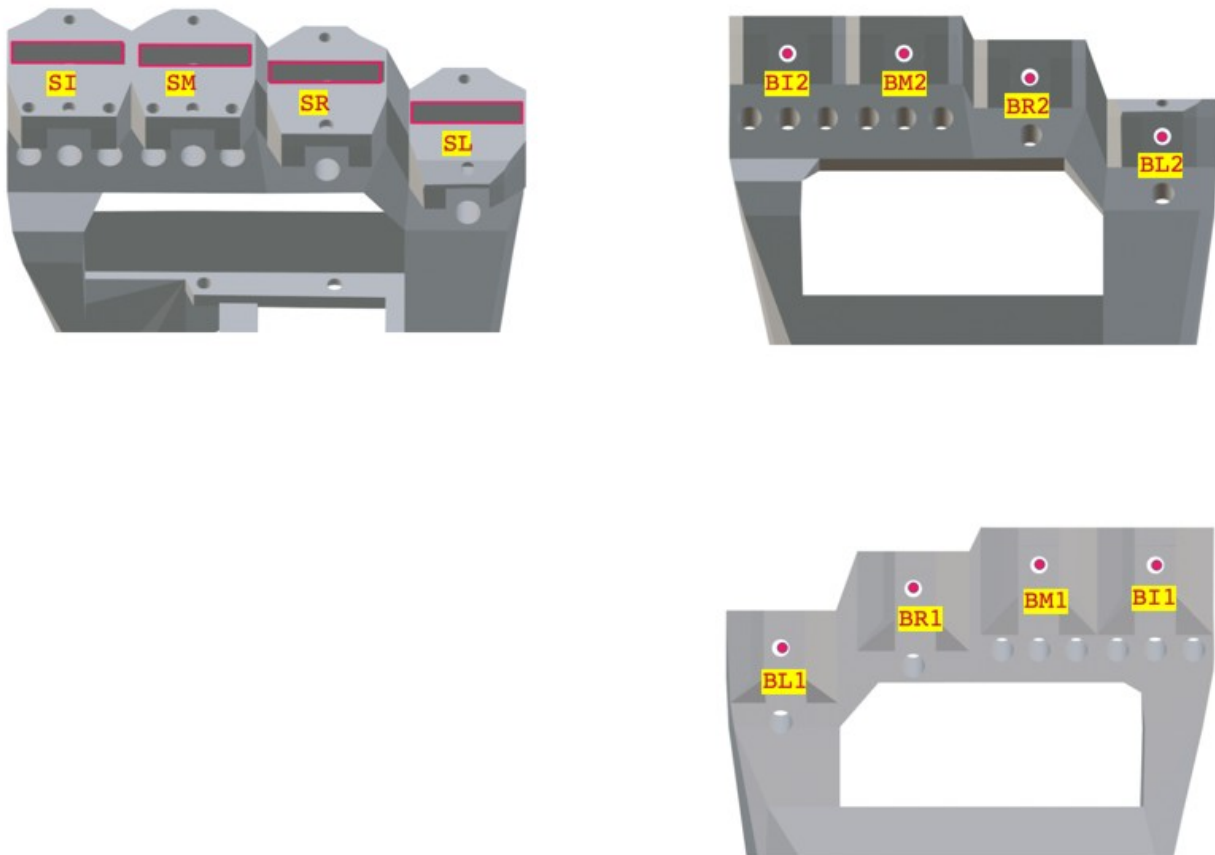


Figure 14.10: Left hand - skeleton - anterior view - upper finger holes and slots

14.2.6 Inch base slot and hole nomenclature

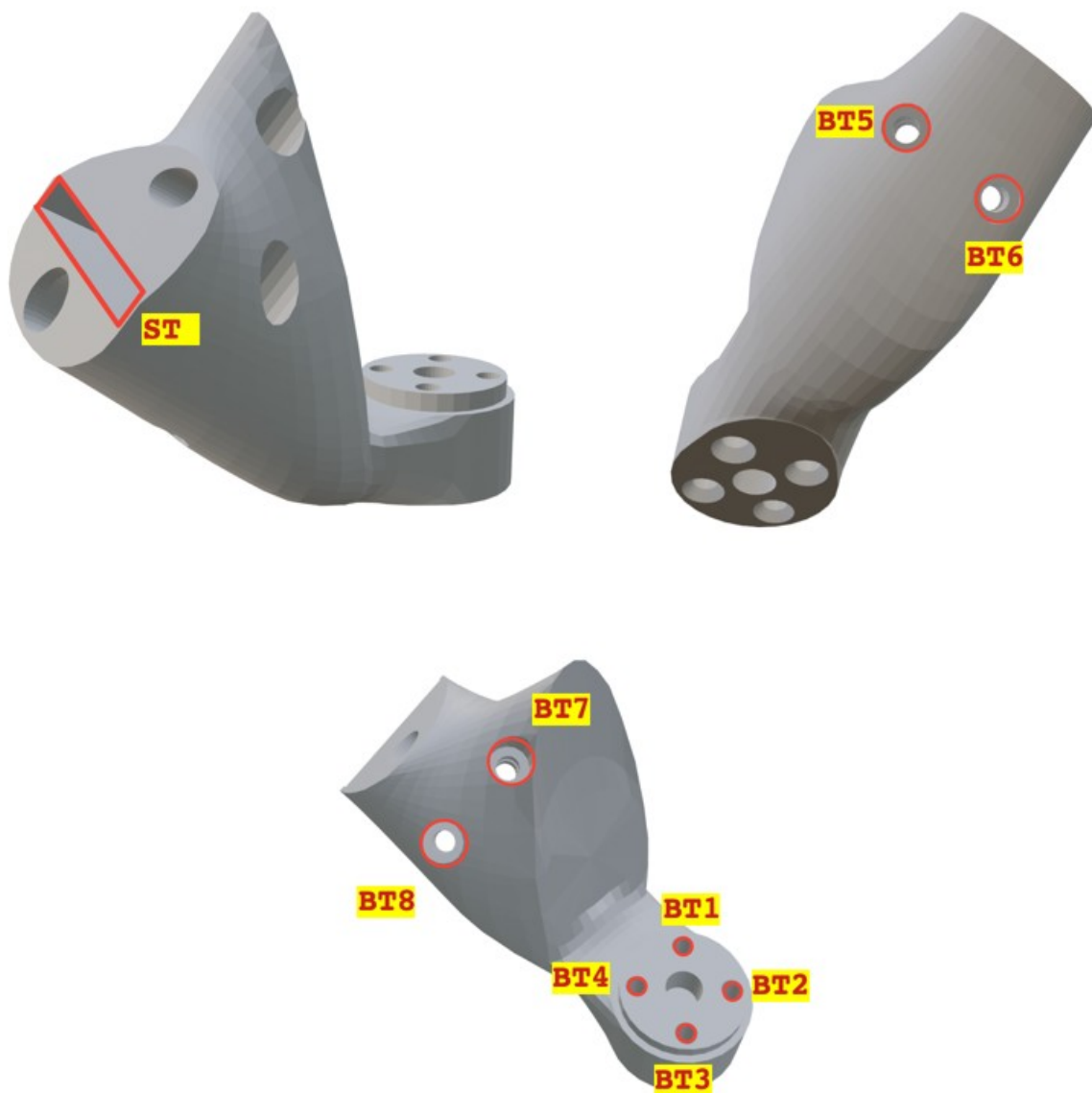


Figure 14.11: Left hand - front view - thumb base holes and slots

14.2.7 Tendon hole nomenclature

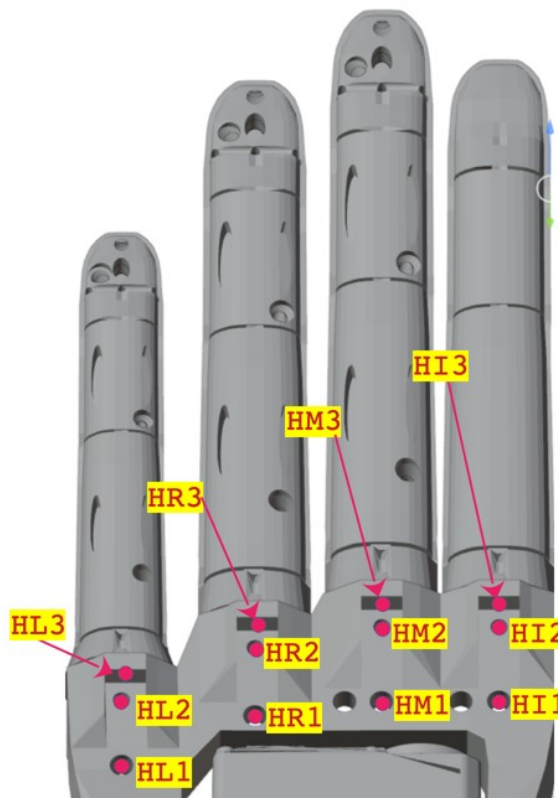


Figure 14.13: Left hand - rear view - tendon holes



Figure 14.12: Left hand - front view - tendon holes



Figure 14.14:
Index finger - hole
distal tendon -
bottom view

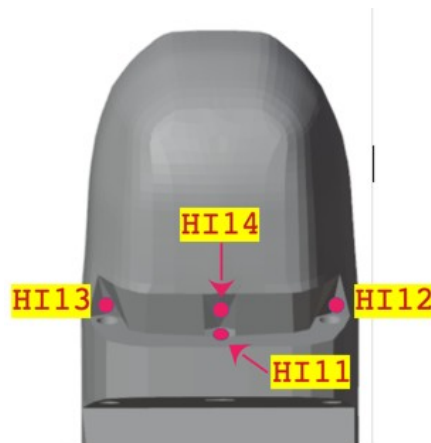


Figure 14.15: Index finger -
distal tendon hole - top view



Figure 14.16: Index finger tendon holes

Only the names for the index finger are shown. The names of the holes for the other fingers can be obtained by replacing the HI prefix with that of the desired finger. Below are the prefixes for the names of the other upper fingers (middle, ring, and little).

- Middle: HM
- Ring: HR
- Little: HL

14.2.8 Thumb tendon hole nomenclature



Figure 14.17: Thumb tendon holes

14.2.9 Winder hole nomenclature

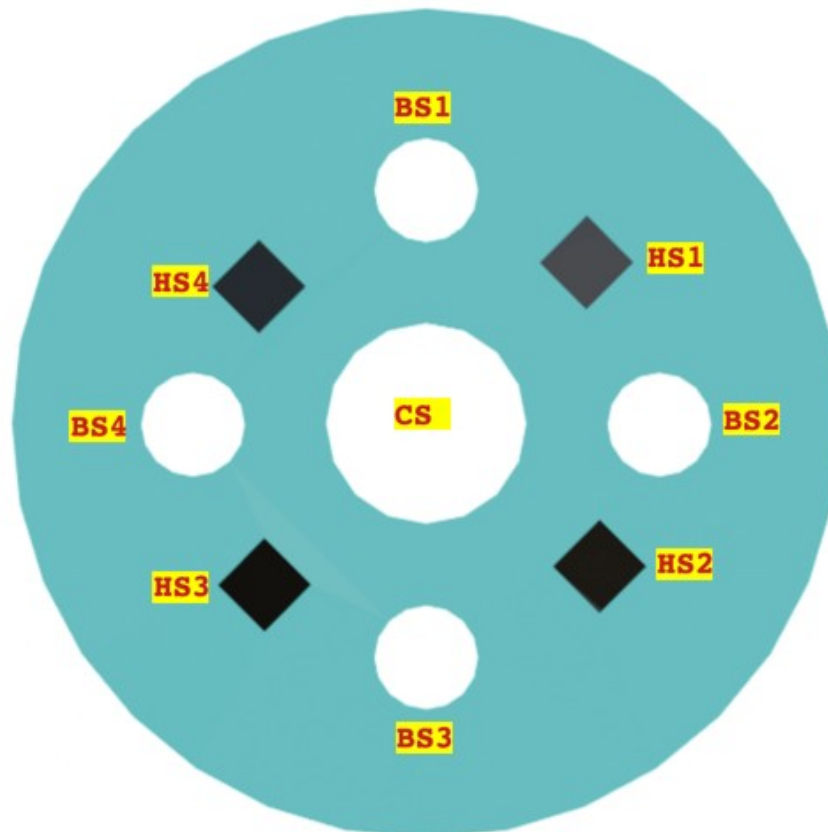


Figure 14.18: winder holes

14.2.10 Palm hole nomenclature

TBD

Figure 14.19: Left hand - palm holes

14.2.11 Covers nomenclature

TBD

14.2.12 Electronic board hole nomenclature

TBD

15 Residual risks to be communicated to users

15.1 General warnings

This section contains the residual risks of the partly completed machinery that **MUST BE COMMUNICATED TO END USERS** in the complete machine manual.

The integrator must incorporate this chapter in its entirety.

15.2 Mechanical risks

- Crushing risk: The robotic fingers exert force up to 20N during gripping. Keep hands/objects away during operation.
- Exposed tendons: Risk of snagging clothing/external objects. Do not touch the tendons during movement.

15.3 Mandatory maintenance (normal wear)

Tendon tension adjustment (calibration) is routine maintenance due to normal wear and tear, required every 5000 cycles (one cycle = one finger closure until resistance, against palm or object)—similar to bicycle brake cable retensioning. Failure to comply voids the warranty coverage. TBD The firmware automatically records cycles; check the app for maintenance alerts.

15.4 Operational limitations

- Do not operate standalone: The partly completed machinery is dangerous without integration on a complete robot with emergency stop and safety barriers.
- Overload:
 - Pinch grasp: 500 g TBD
 - Power grasp: 5 kg TBD
 - Lift: 8 kg TBD
- Environment: Temperature 5-40°C, humidity <80% non-condensing.

15.5 Operator training

Train operators on: risks listed above, tendon calibration procedure, energy isolation (Chap 10.5)

The integrator must translate and include this chapter in the end-user manual of the complete machine.